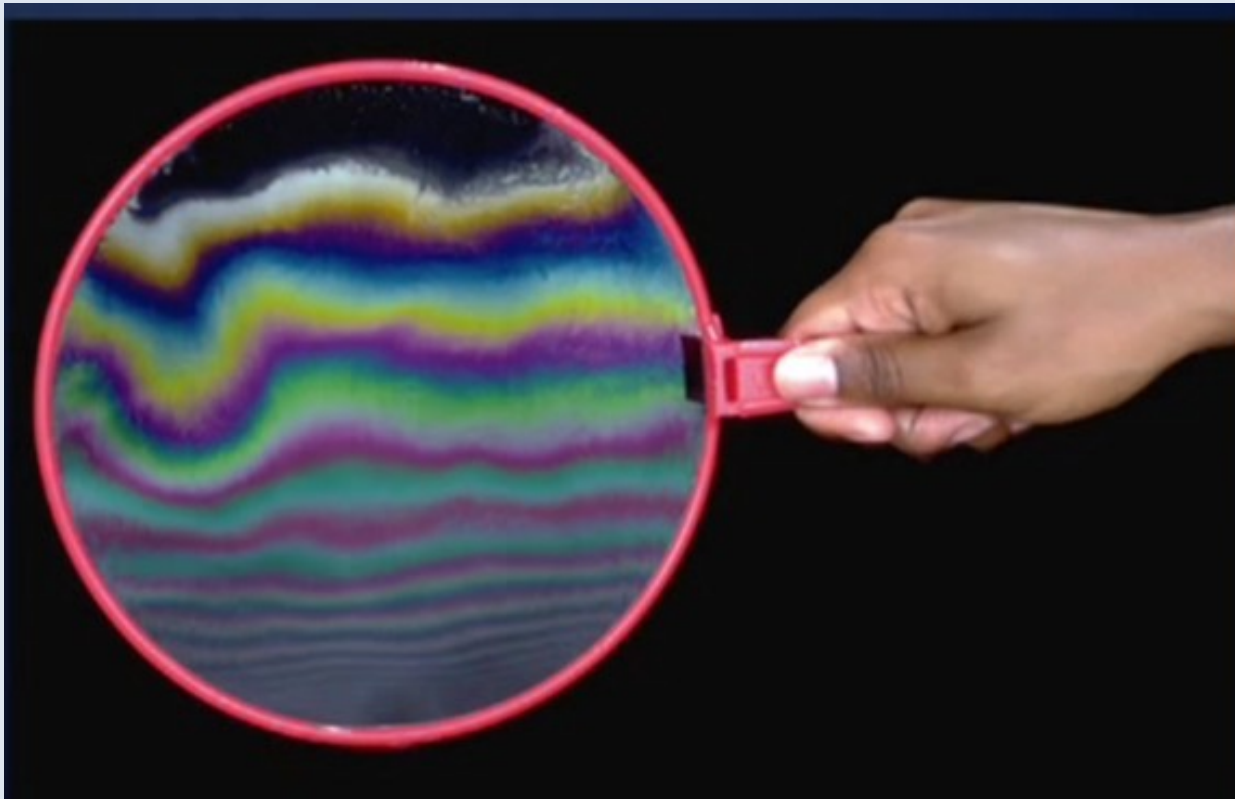


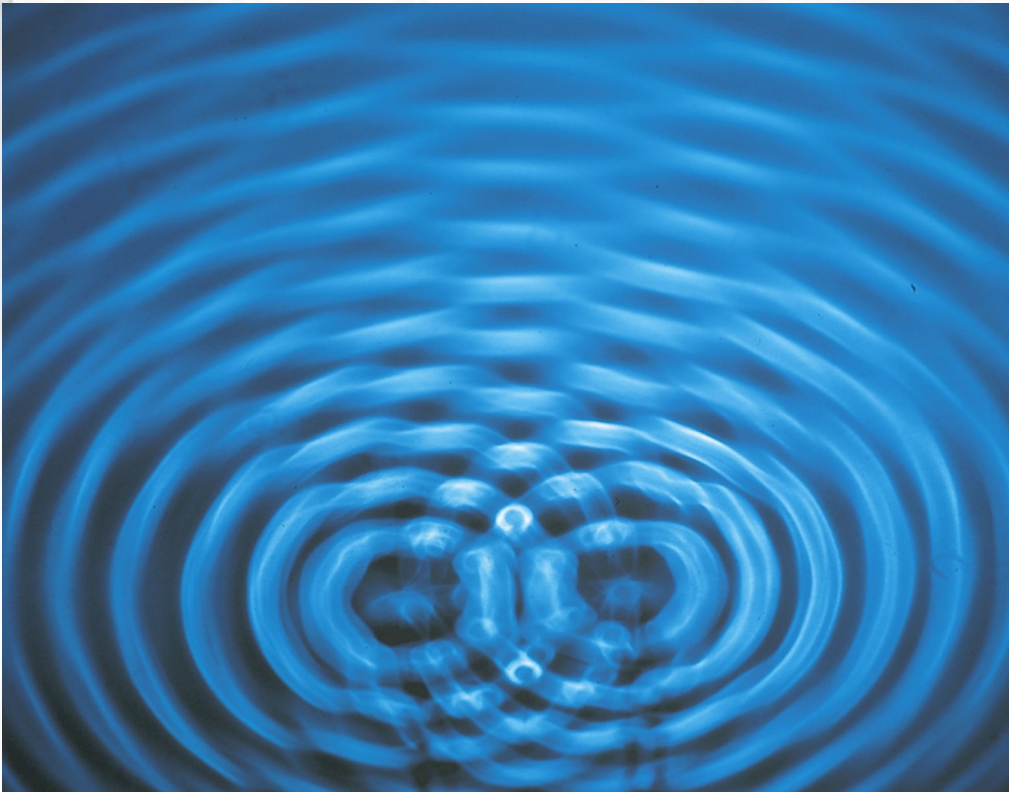
Wave Optics

Interference



CHAPTER 3 : interference, standing waves, diffraction

- *Interference* occurs whenever two or more waves overlap.



in physics,
the net effect of the combination
of two or more wave trains
moving on intersecting or
coincident paths. The effect is
that of the addition of the
amplitudes of the individual
waves at each point affected
by more than one wave.

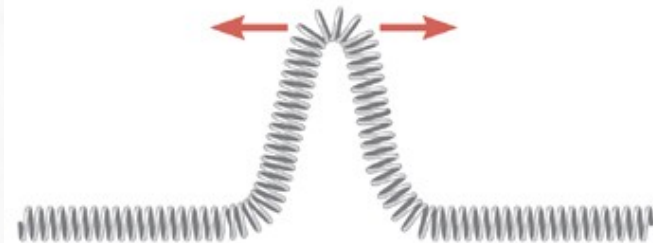
See applets from website

17.1 The Principle of Linear Superposition

When the pulses merge, the Slinky assumes a shape that is the sum of the shapes of the individual pulses.



(a) Overlap begins



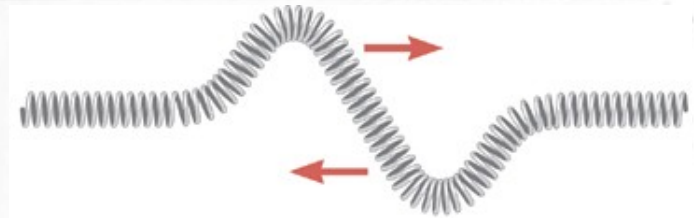
(b) Total overlap; the Slinky has twice the height of either pulse



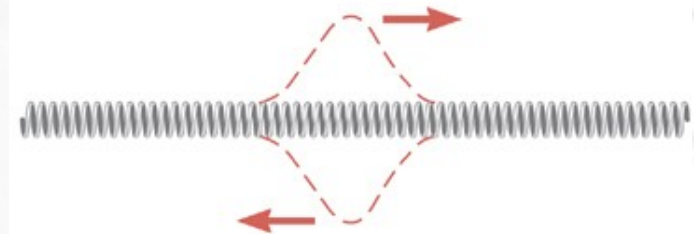
(c) The receding pulses

17.1 The Principle of Linear Superposition

When the pulses merge, the Slinky assumes a shape that is the sum of the shapes of the individual pulses.



(a) Overlap begins



(b) Total overlap



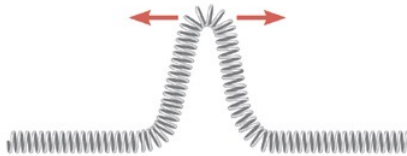
(c) The receding pulses

THE PRINCIPLE OF LINEAR SUPERPOSITION

When two or more waves are present simultaneously at the same place, the resultant disturbance is the sum of the disturbances from the individual waves.



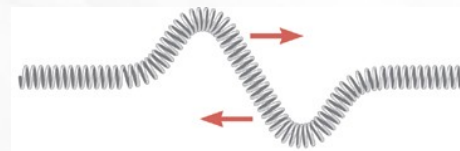
(a) Overlap begins



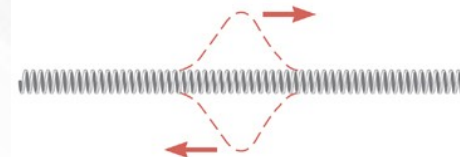
(b) Total overlap; the Slinky has twice the height of either pulse



(c) The receding pulses



(a) Overlap begins

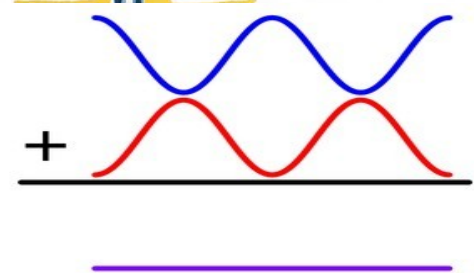
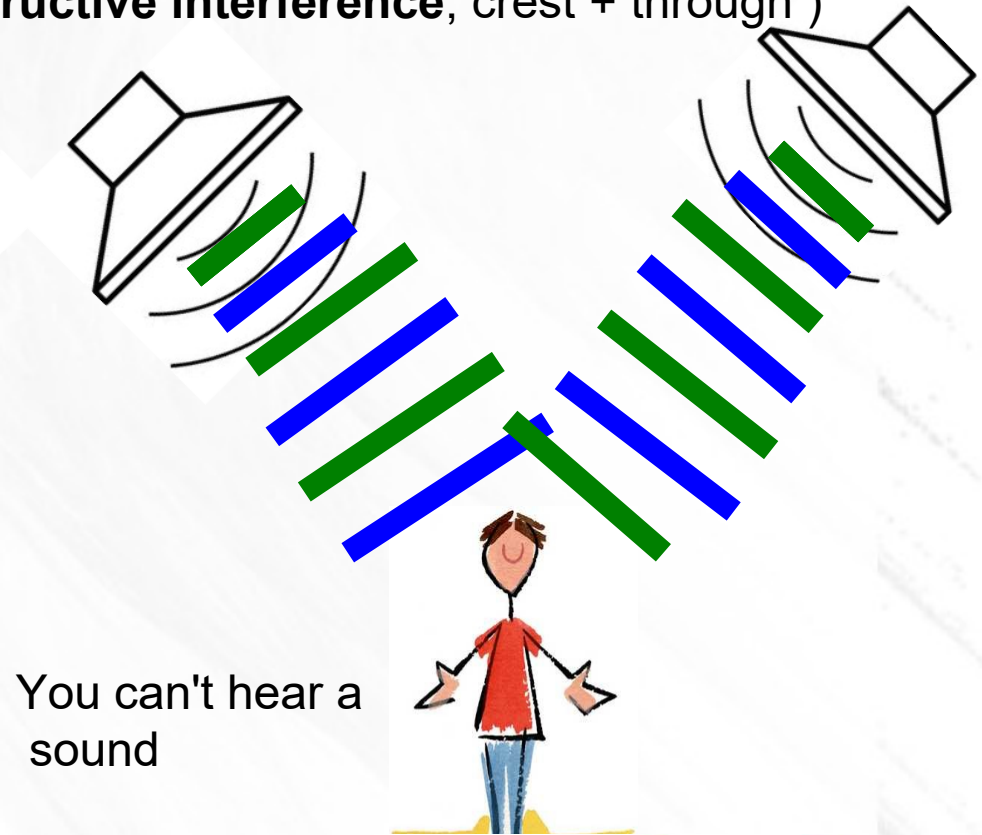
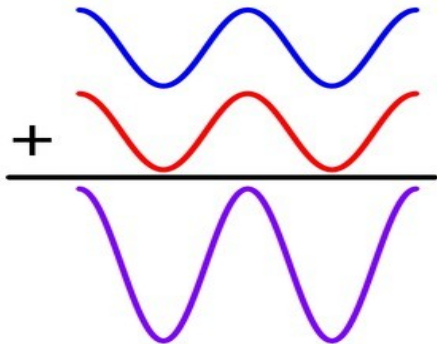
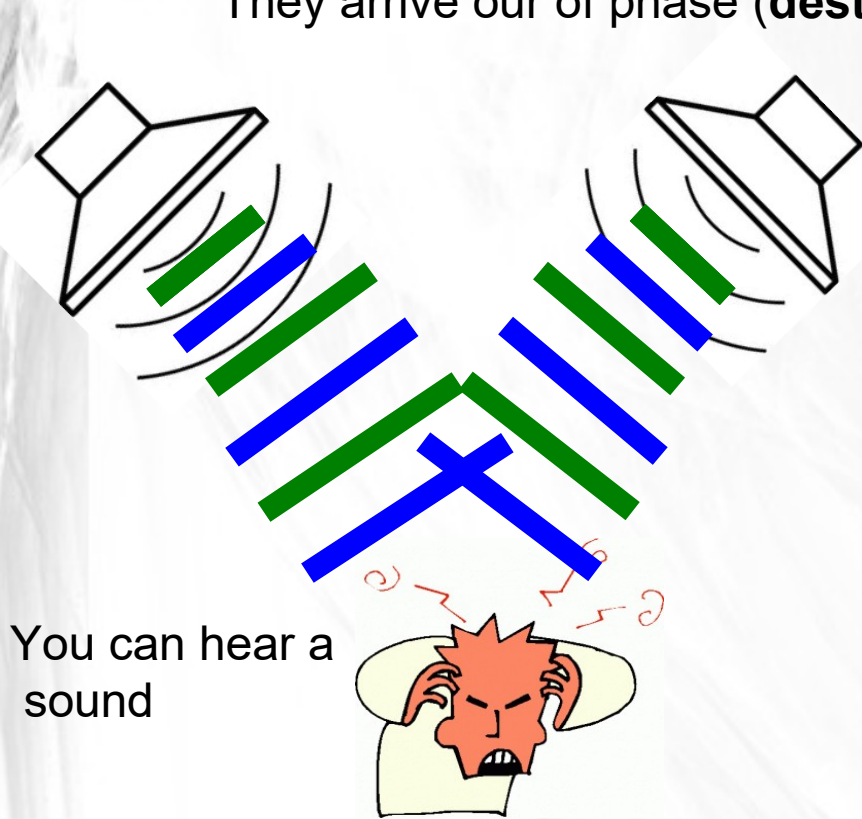


(b) Total overlap



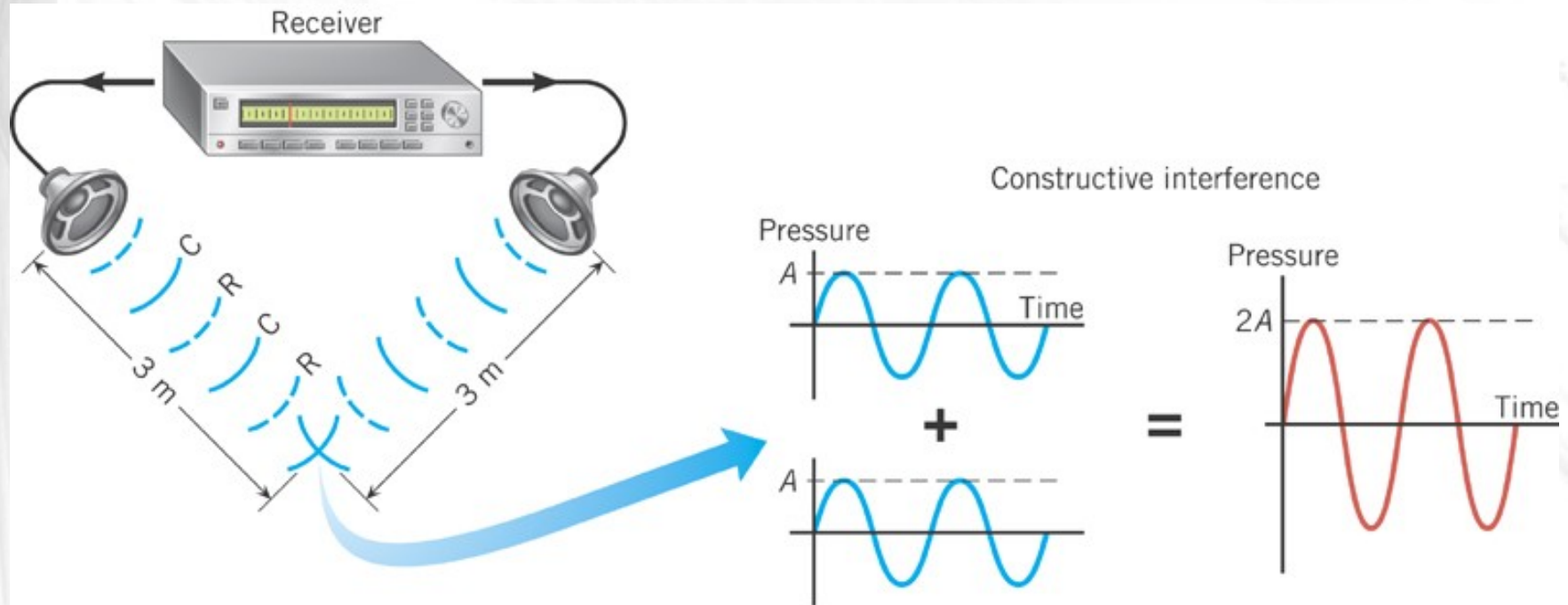
(c) The receding pulses

Interference of sound. In a concert room of bad quality, you can't hear anything at some places. Depending if the waves arrive in phase (**constructive interference**, the crests add up) or They arrive out of phase (**destructive interference**, crest + trough)



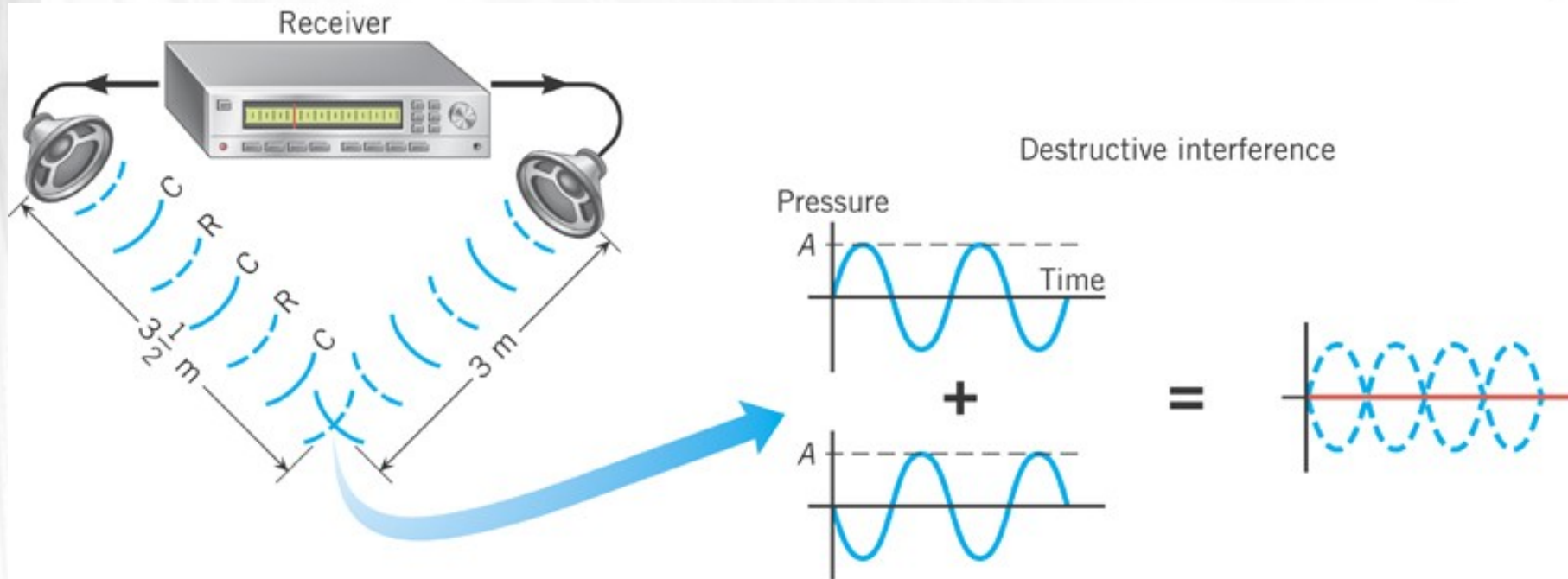
17.2 Constructive and Destructive Interference of Sound Waves

When two waves always meet condensation-to-condensation and rarefaction-to-rarefaction, they are said to be ***exactly in phase*** and to exhibit ***constructive interference***.



17.2 Constructive and Destructive Interference of Sound Waves

When two waves always meet condensation-to-rarefaction, they are said to be **exactly out of phase** and to exhibit **destructive interference**.



Constructive Interference

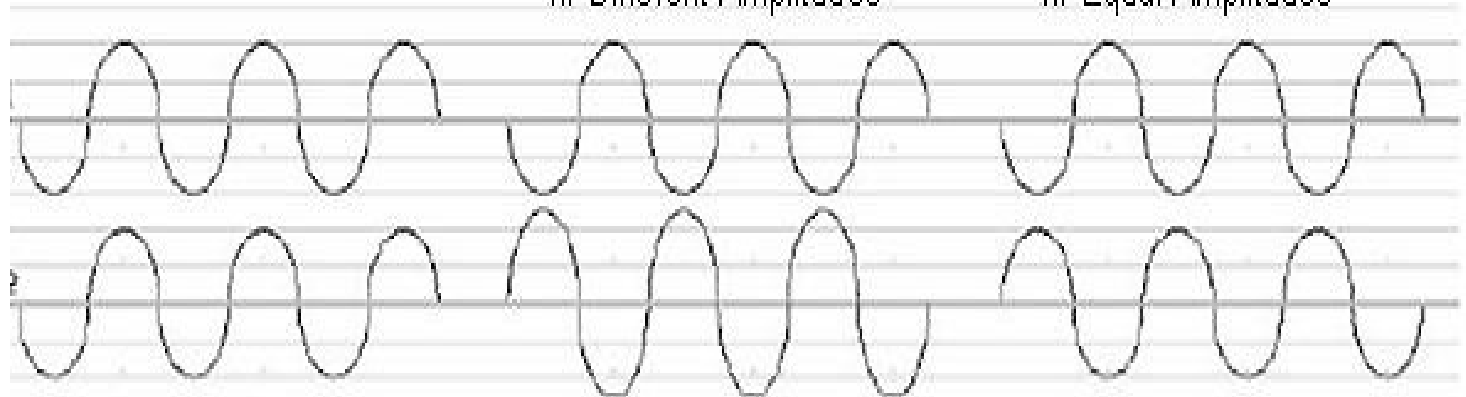
Destructive Interference
w/ Different Amplitudes

Destructive Interference
w/ Equal Amplitudes

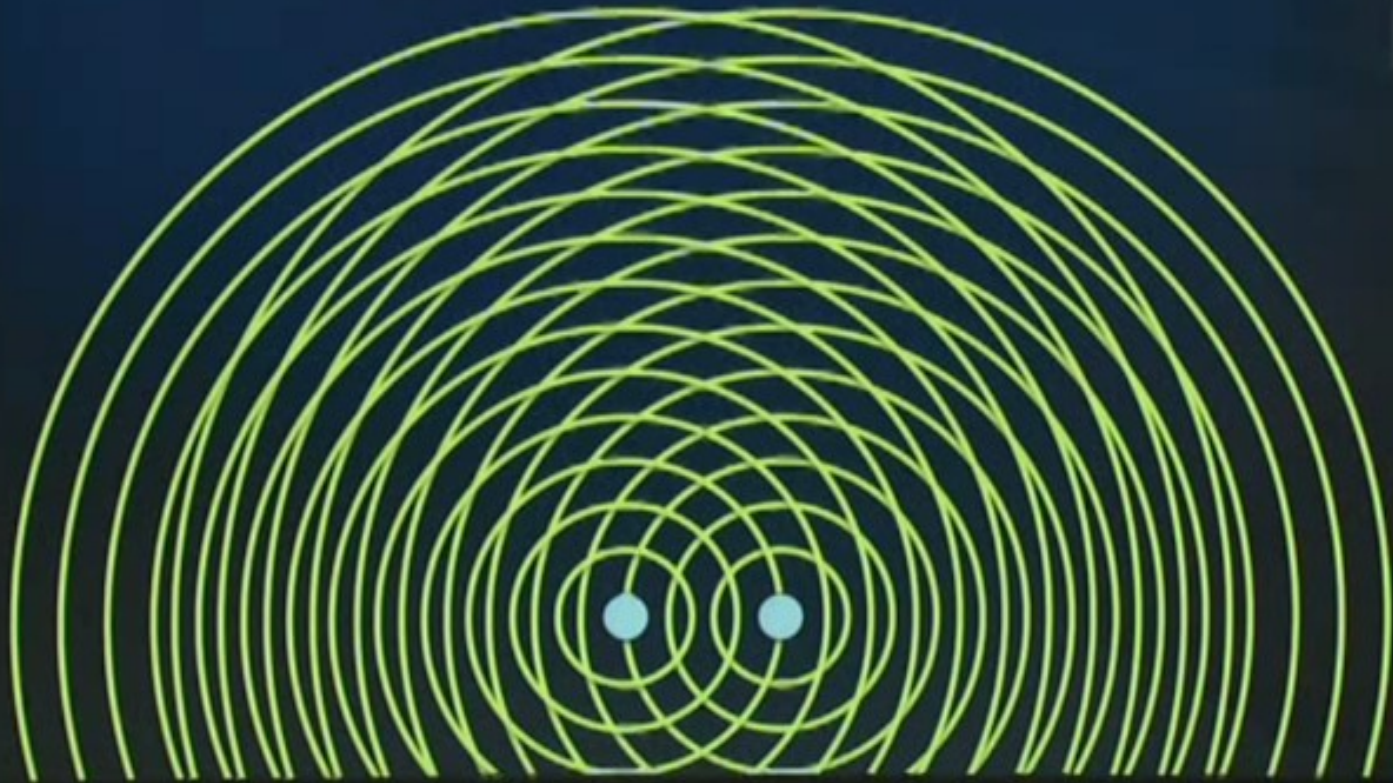
Light Wave 1

Light Wave 2

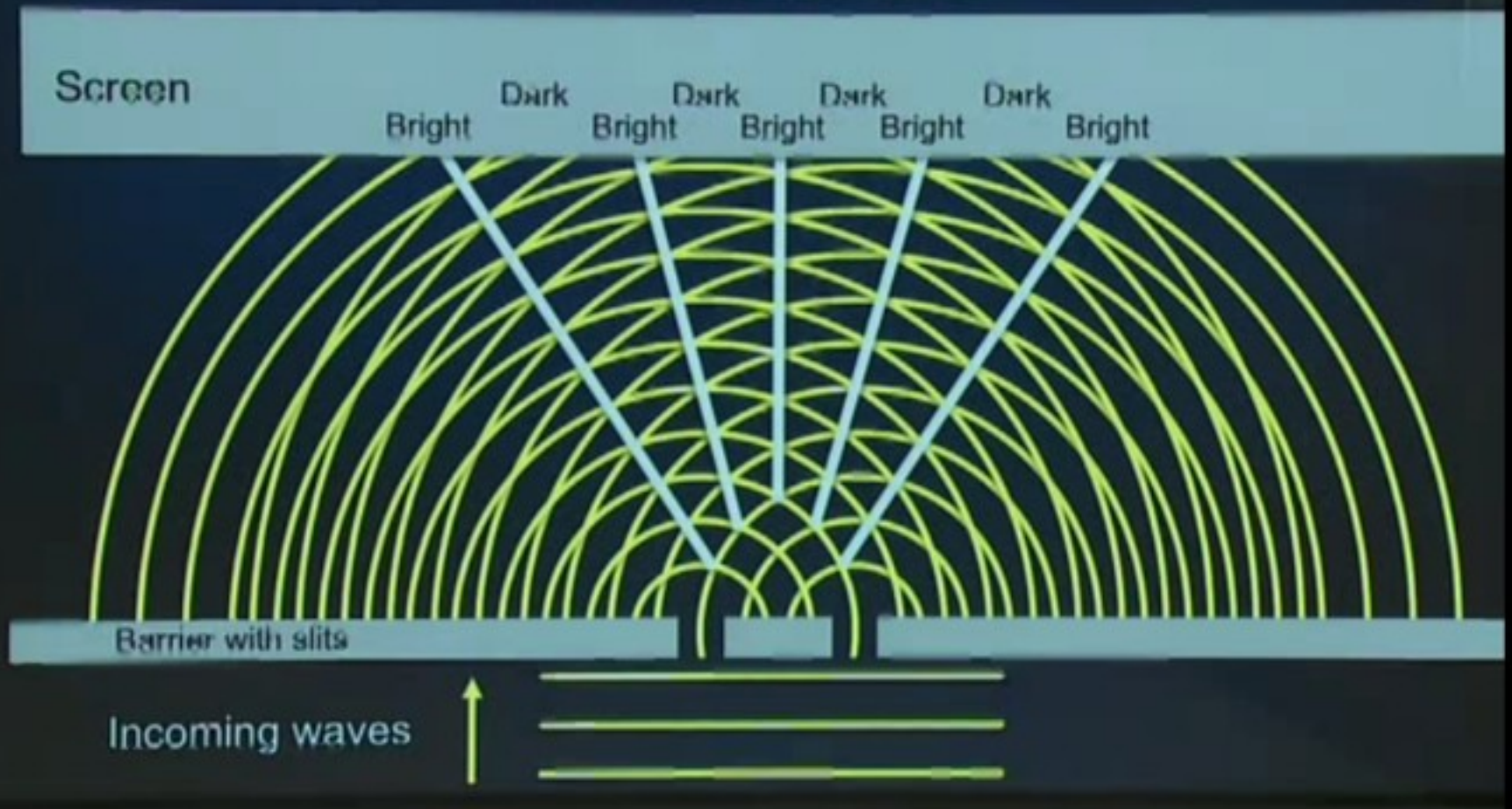
Light Wave 1 + 2



2-Source Interference



2-Slit Interference



2-Slit Interference

Constructive interference:

Path difference $r_2 - r_1$ is a multiple of the wavelength λ
 $\Rightarrow d \sin \theta = m\lambda$

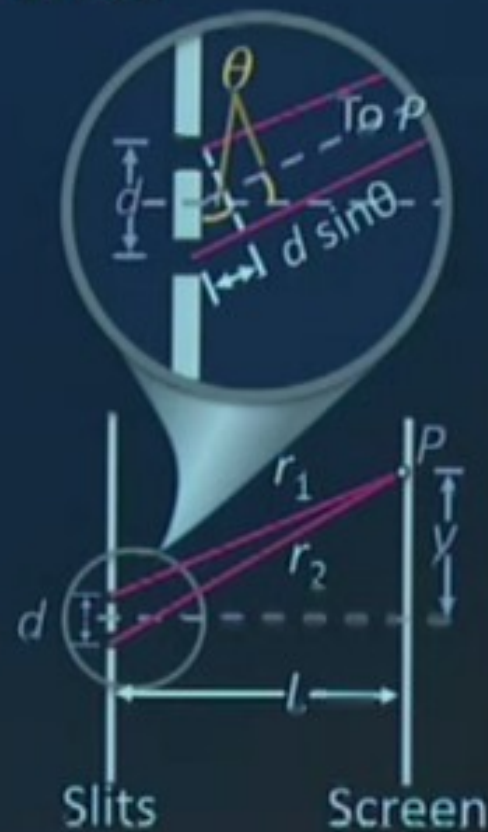
For small θ , $\sin \theta \approx \tan \theta \approx y/L$

So bright fringes when $d \frac{y}{L} = m\lambda$
 $\Rightarrow y = m \frac{\lambda L}{d}$

Destructive interference:

Path difference $r_2 - r_1$ is an odd multiple of a half-wavelength

Dark fringes at $y = \left(m + \frac{1}{2}\right) \frac{\lambda L}{d}$



Wavelength of Light

Bright fringes at $y = m \frac{\lambda L}{d}$

Dark fringes at $y = \left(m + \frac{1}{2}\right) \frac{\lambda L}{d}$

Do the numbers:

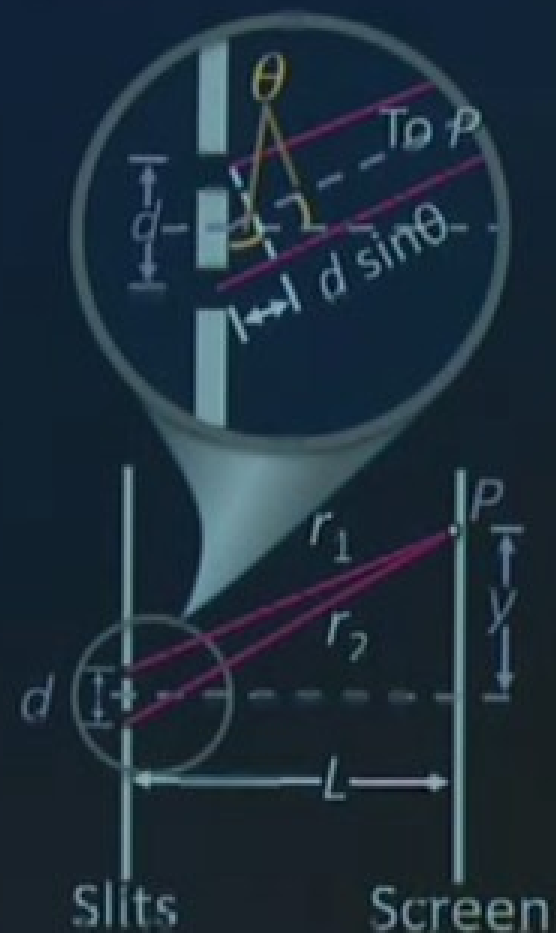
Solve bright-fringe equation for λ :

First fringe ($m = 1$): $\lambda = \frac{yd}{L}$

$$\Rightarrow \lambda = \frac{yd}{L} = \frac{(2.6 \text{ mm})(0.25 \text{ mm})}{(1000 \text{ mm})}$$

$$= 0.00065 \text{ mm}$$

$$= 650 \text{ nm}$$



Multiple Slits

- For maximum:

$$d \sin \theta = m \lambda$$

- For minimum:

$$d \sin \theta = \frac{m}{N} \lambda$$

m not a multiple of N

2 slit



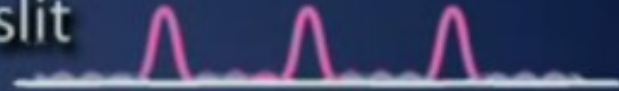
3 slit



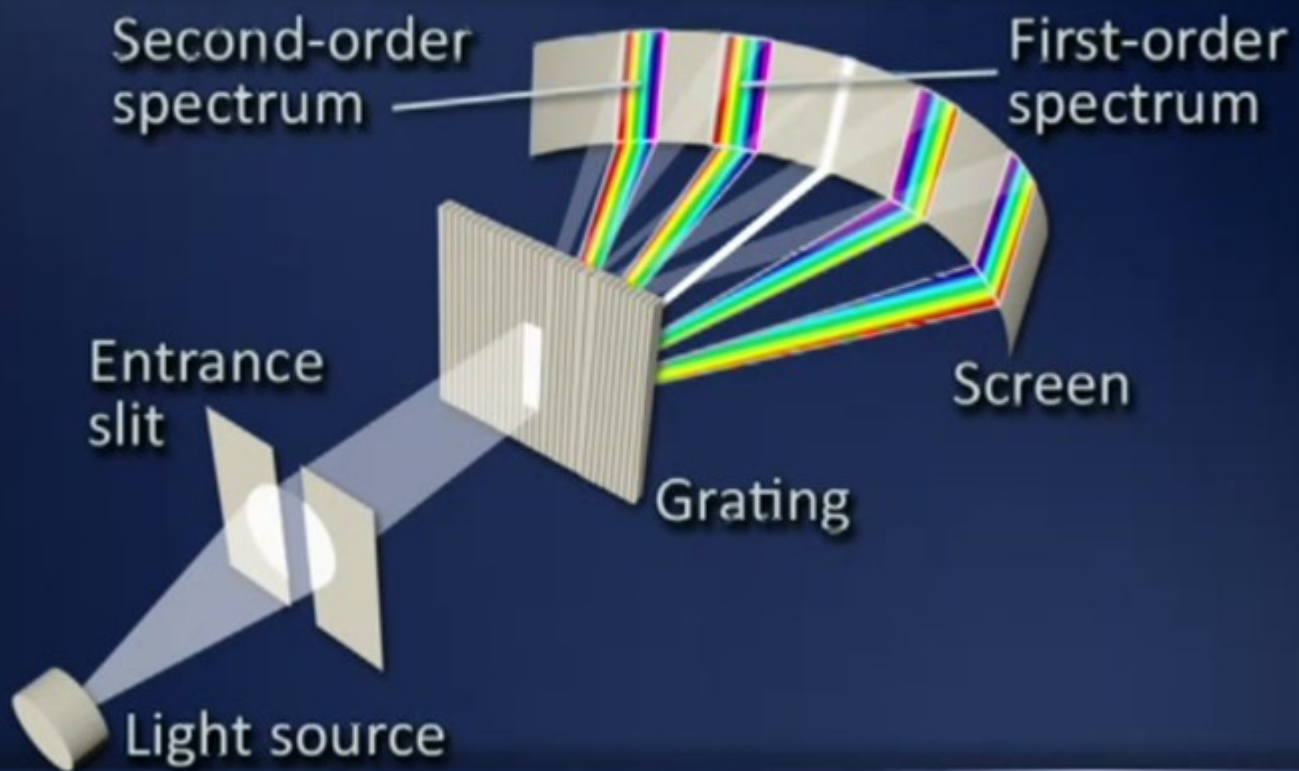
4 slit



5 slit

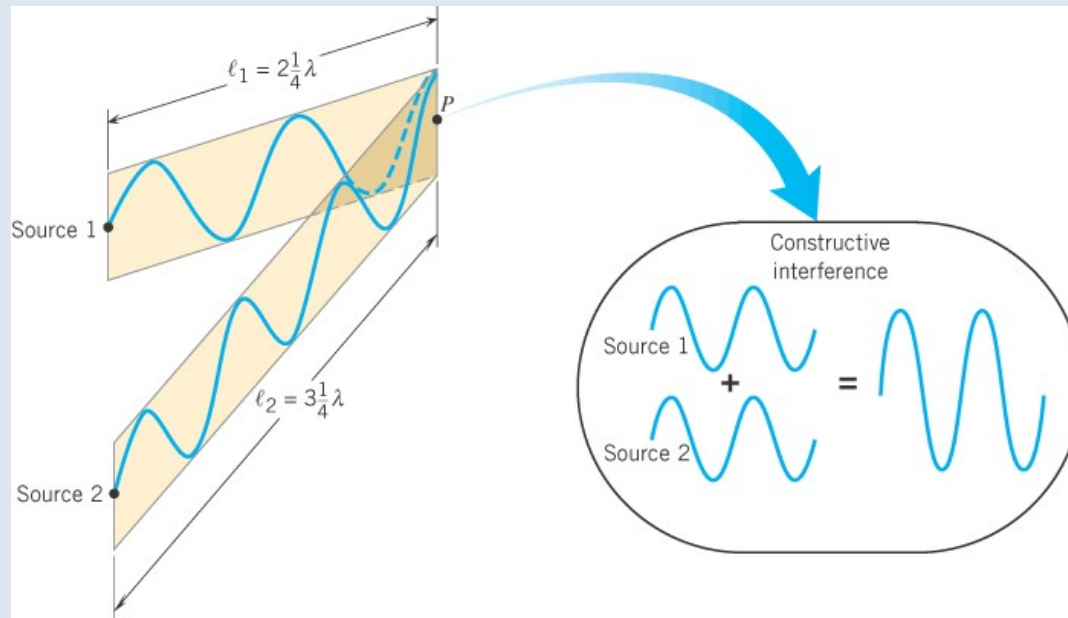


Analyzing Light: A Grating Spectrometer



27.1 The Principle of Linear Superposition

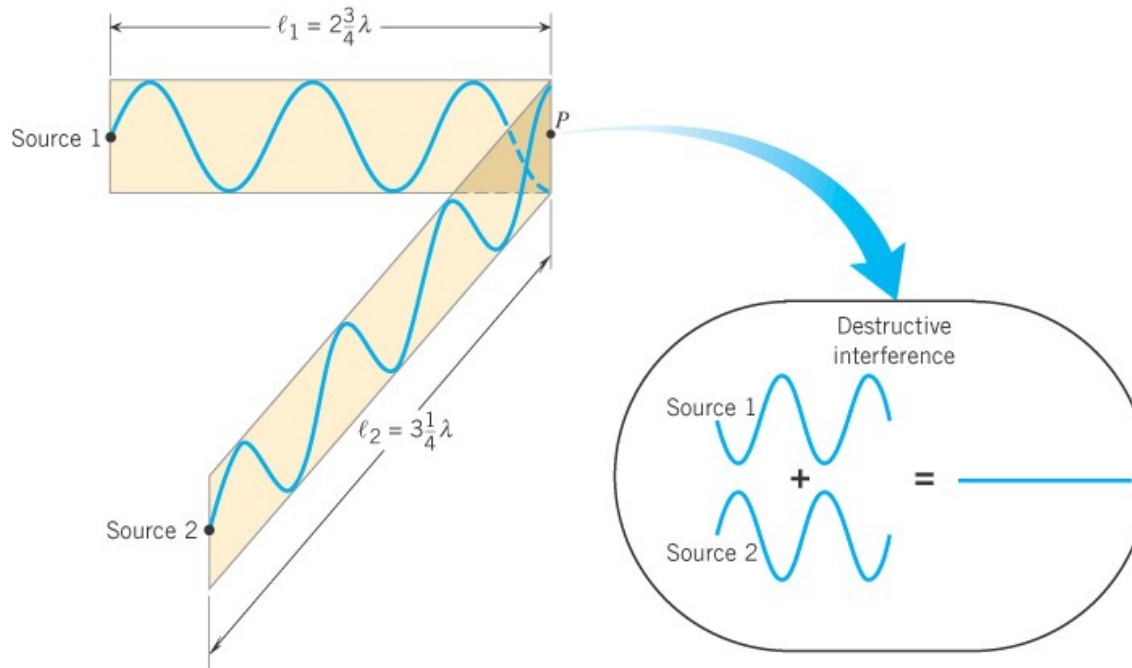
When two or more light waves pass through a given point, their electric fields combine according to the principle of superposition.



The waves emitted by the sources start out in phase and arrive at point P in phase, leading to **constructive interference**.

$$\ell_2 - \ell_1 = m\lambda \quad m = 0, 1, 2, 3, \dots$$

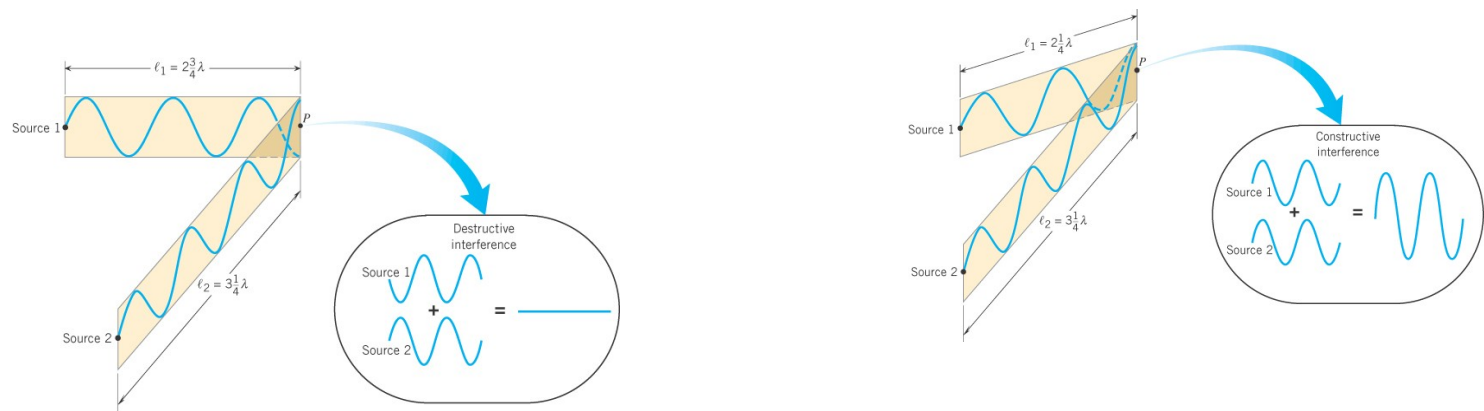
27.1 The Principle of Linear Superposition



The waves emitted by the sources start out in phase and arrive at point P out of phase, leading to ***destructive interference***.

$$\ell_2 - \ell_1 = \left(m + \frac{1}{2}\right)\lambda \quad m = 0, 1, 2, 3, \dots$$

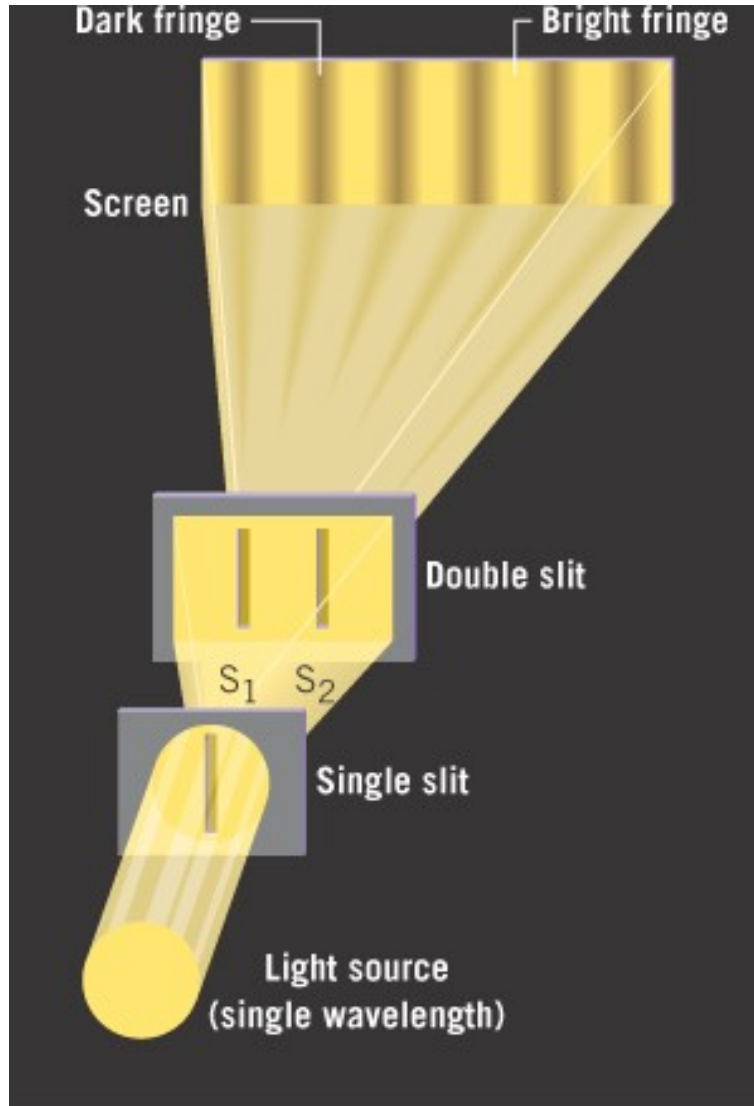
27.1 The Principle of Linear Superposition



If constructive or destructive interference is to continue occurring at a point, the sources of the waves must be **coherent sources**.

Two sources are coherent if the waves they emit maintain a constant phase relation.

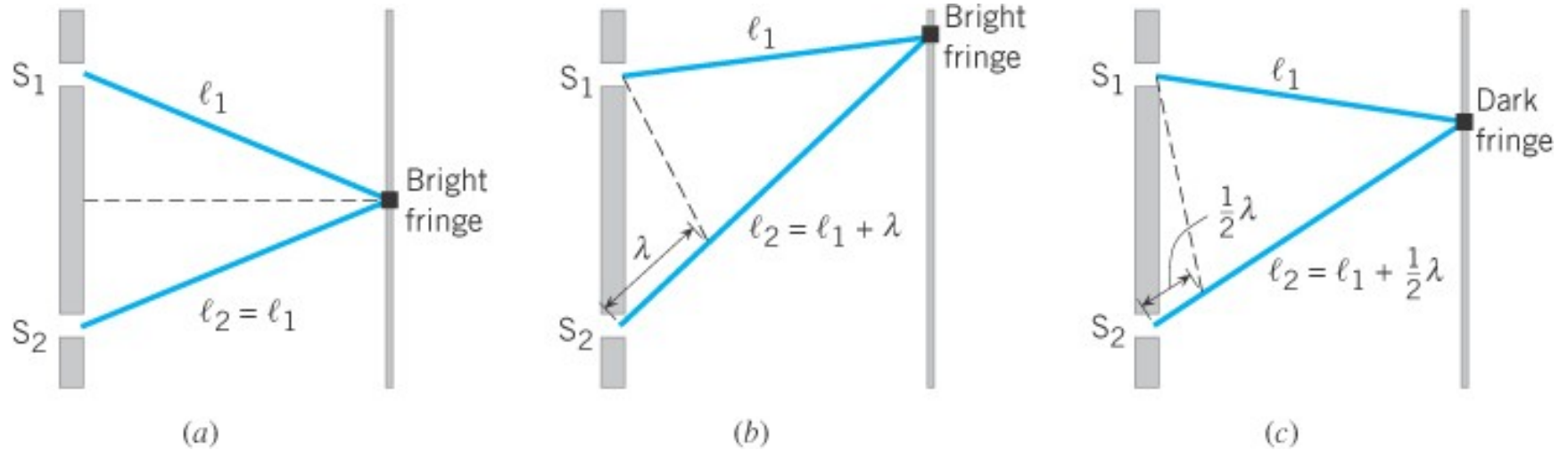
27.2 Young's Double Slit Experiment



In Young's experiment, two slits act as coherent sources of light.

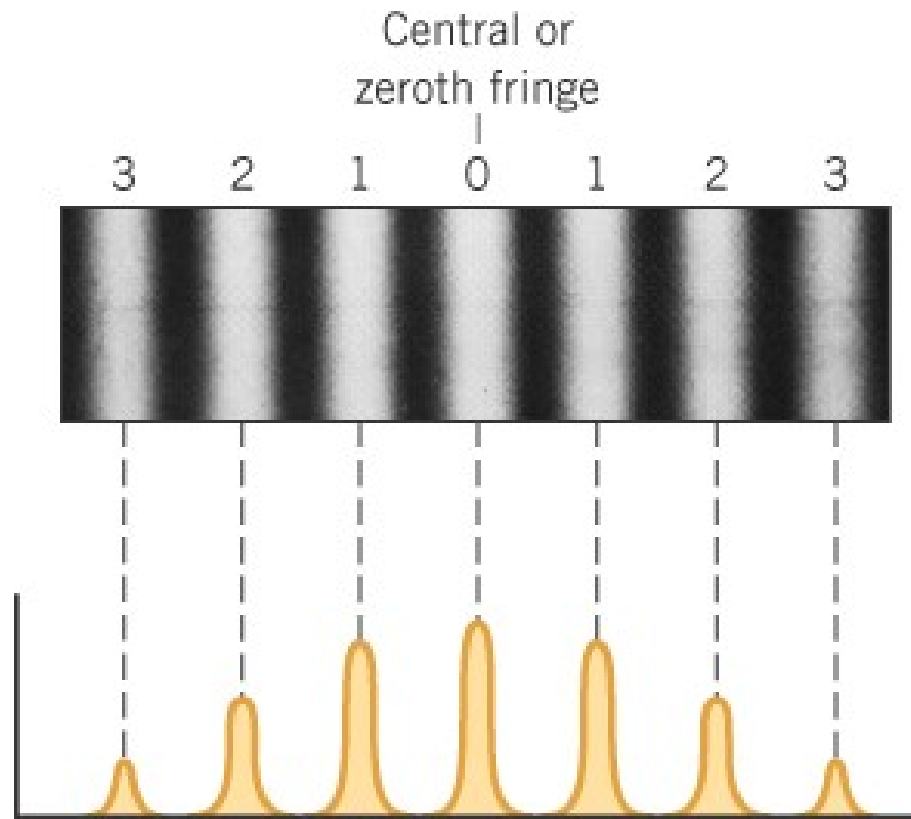
Light waves from these slits interfere constructively and destructively on the screen.

27.2 Young's Double Slit Experiment

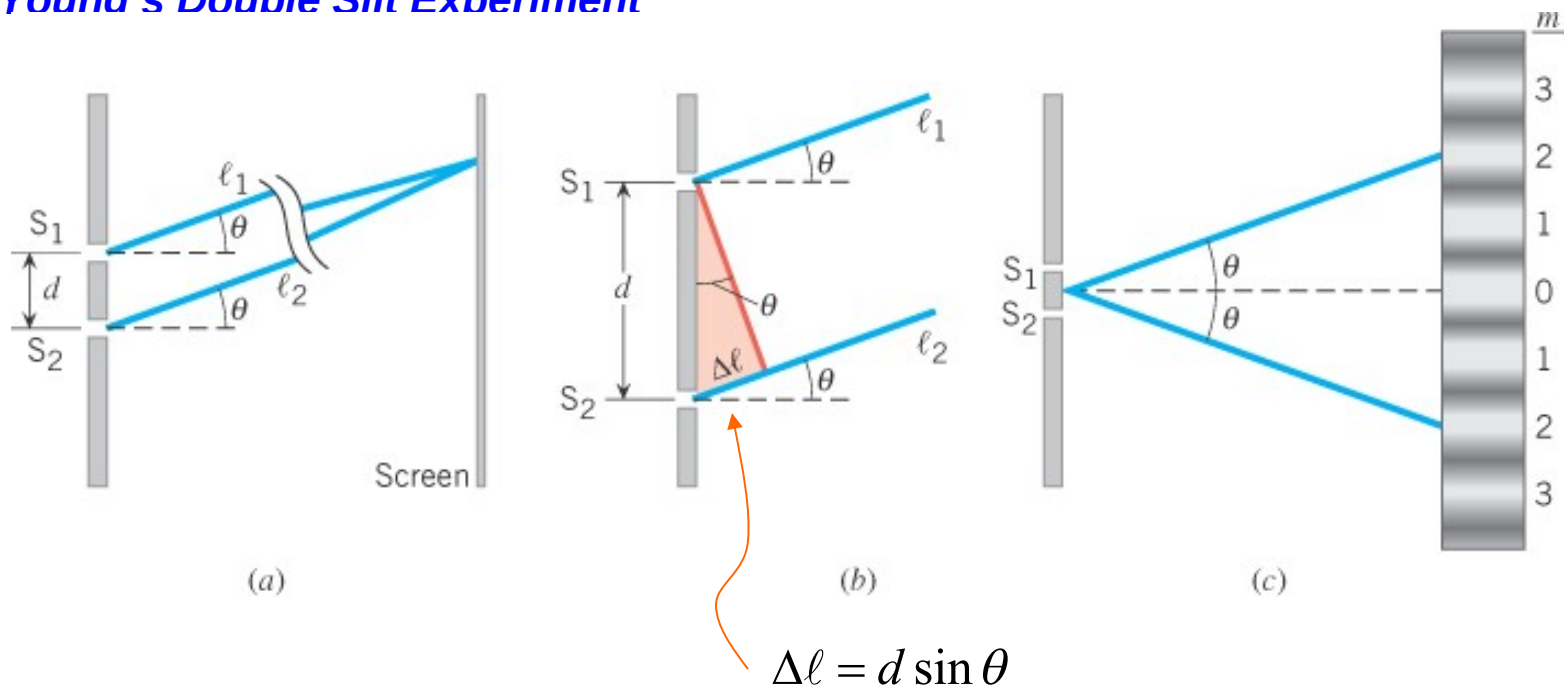


The waves coming from the slits interfere constructively or destructively, depending on the difference in distances between the slits and the screen.

27.2 Young's Double Slit Experiment



27.2 Young's Double Slit Experiment



**Bright fringes
of a double-slit**

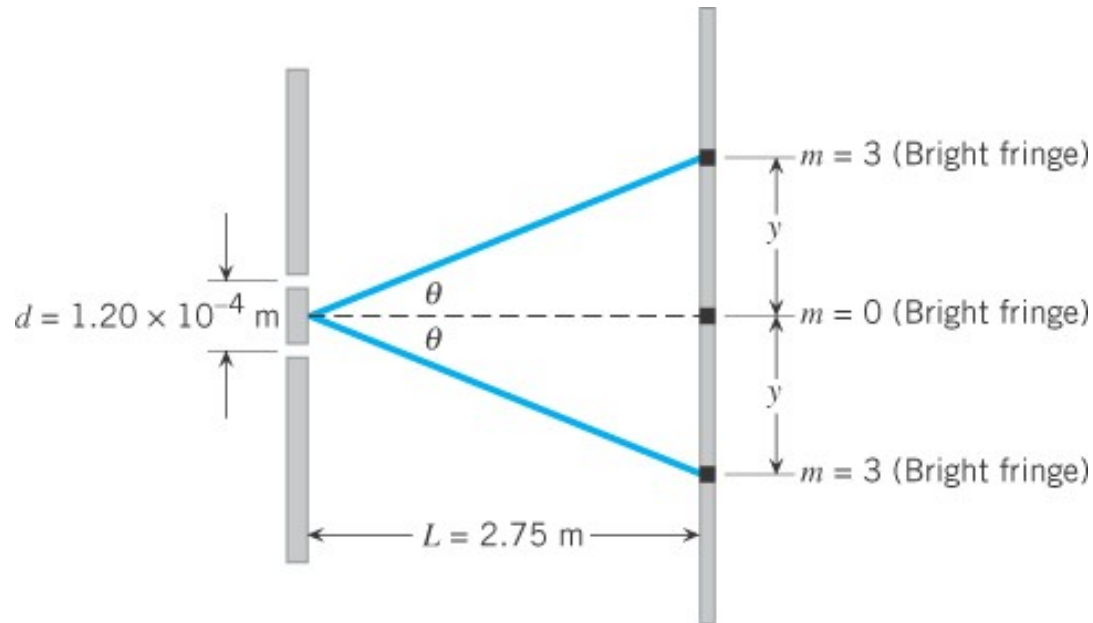
$$\sin \theta = m \frac{\lambda}{d} \quad m = 0, 1, 2, 3, \dots$$

**Dark fringes
of a double-slit**

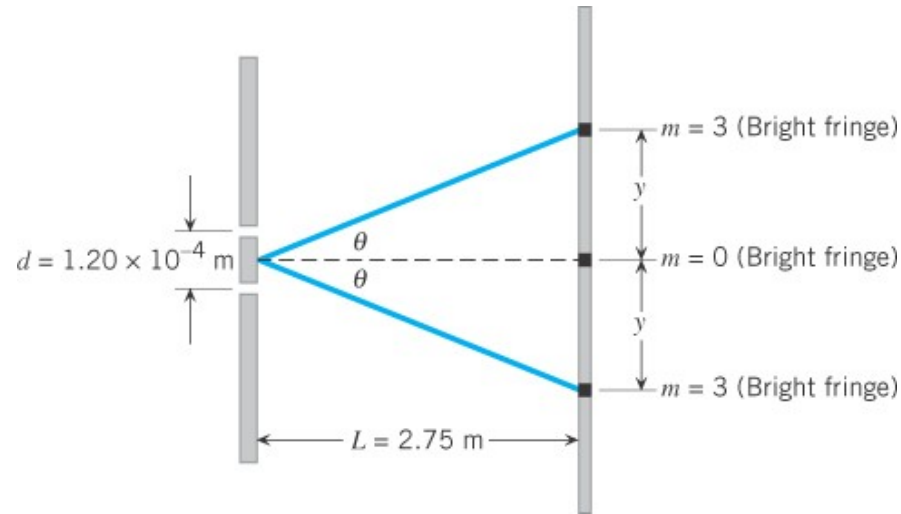
$$\sin \theta = \left(m + \frac{1}{2}\right) \frac{\lambda}{d} \quad m = 0, 1, 2, 3, \dots$$

Example 1 Young's Double-Slit Experiment

Red light (664 nm) is used in Young's experiment with slits separated by 0.000120 m. The screen is located a distance 2.75 m from the slits. Find the distance on the screen between the central bright fringe and the third-order bright fringe.



27.2 Young's Double Slit Experiment

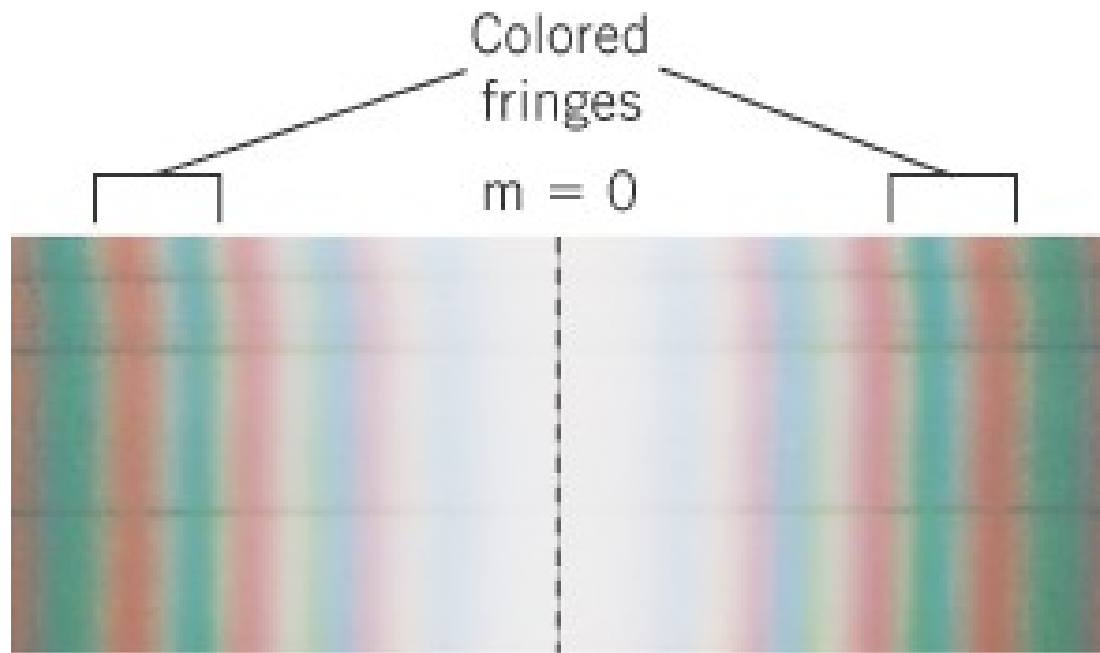


$$\theta = \sin^{-1}\left(m \frac{\lambda}{d}\right) = \sin^{-1}\left(3 \frac{664 \times 10^{-9} \text{ m}}{1.20 \times 10^{-4} \text{ m}}\right) = 0.951^\circ$$

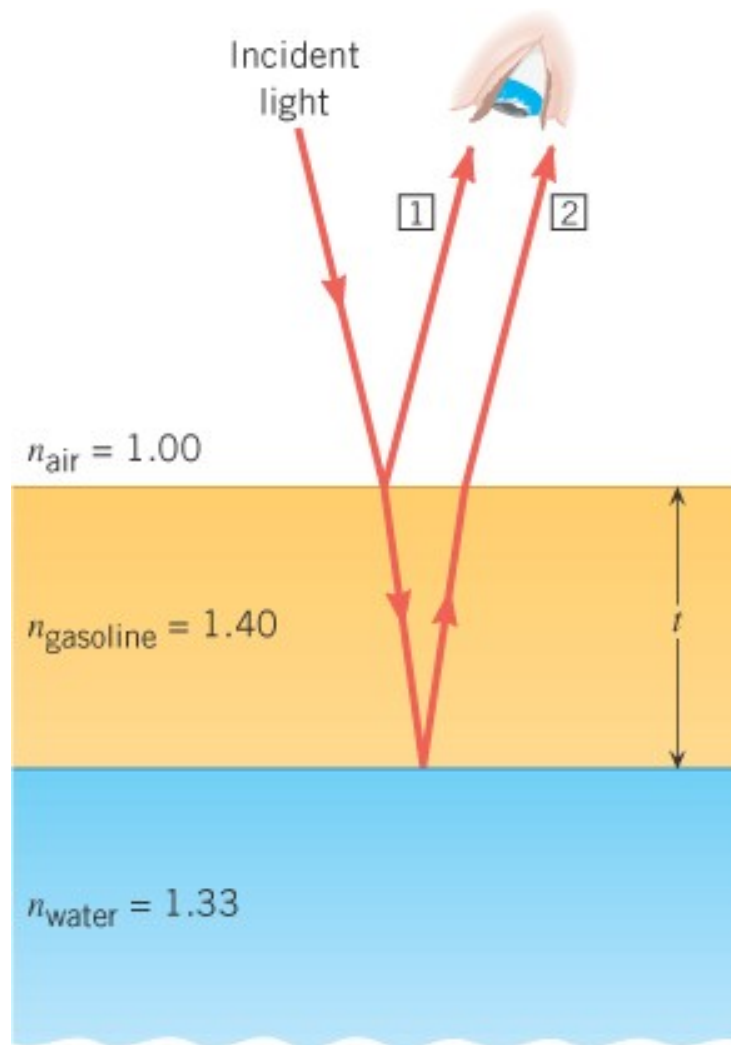
$$y = L \tan \theta = (2.75 \text{ m}) \tan(0.951^\circ) = 0.0456 \text{ m}$$

Conceptual Example 2 White Light and Young's Experiment

The figure shows a photograph that illustrates the kind of interference fringes that can result when white light is used in Young's experiment. Why does Young's experiment separate white light into its constituent colors? In any group of colored fringes, such as the two singled out, why is red farther out from the central fringe than green is? Why is the central fringe white?



27.3 Thin Film Interference



Because of reflection and refraction, two light waves enter the eye when light shines on a thin film of gasoline floating on a thick layer of water.

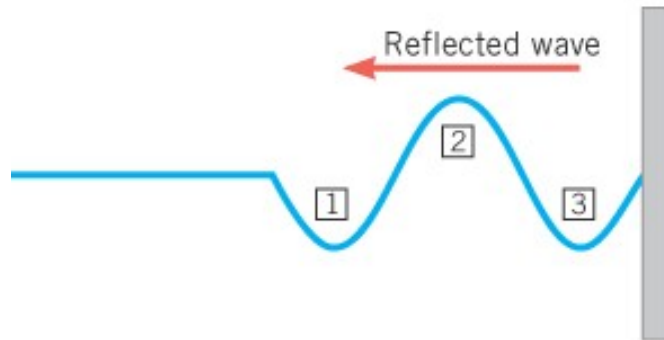
Because of the extra distance traveled, there can be interference between the two waves.

$$\lambda_{\text{film}} = \frac{\lambda_{\text{vacuum}}}{n}$$

27.3 Thin Film Interference



(a)



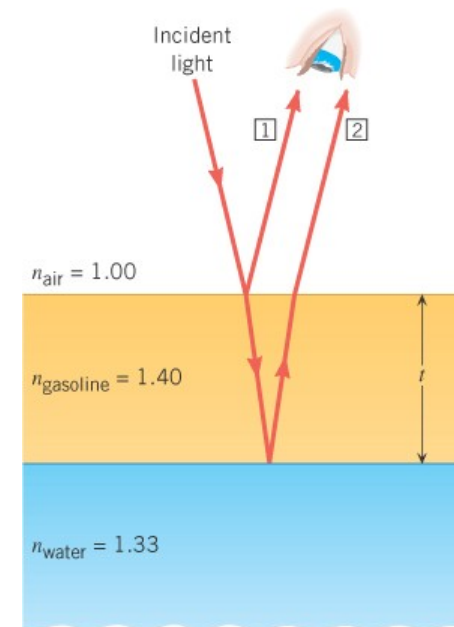
(b)

When light travels through a material with a smaller refractive index towards a material with a larger refractive index, reflection at the boundary occurs along with a phase change that is equivalent to one-half of a wavelength in the film.

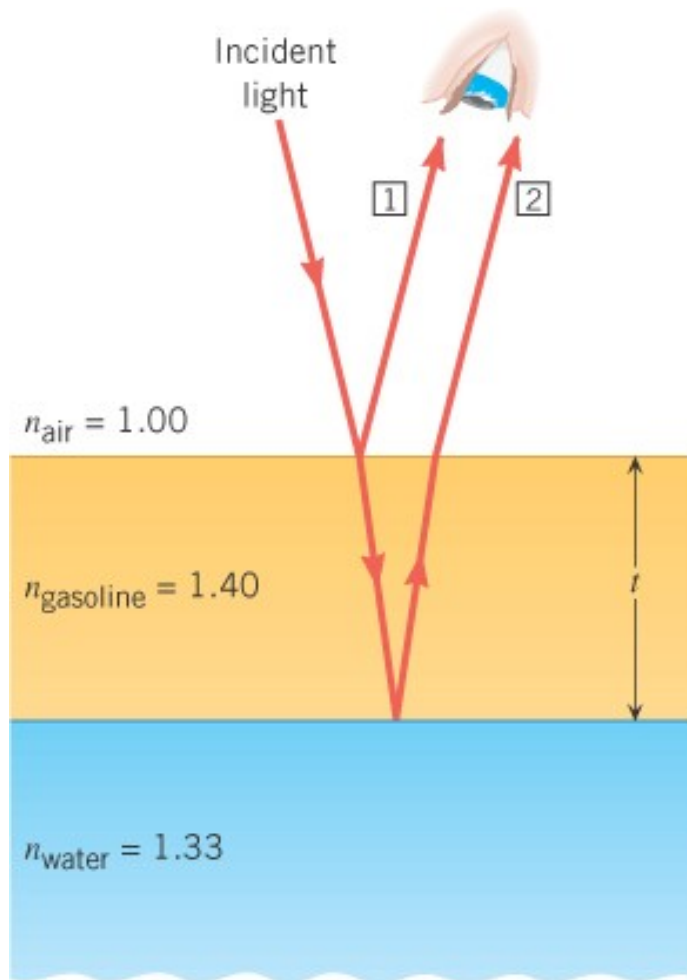
When light travels from a larger towards a smaller refractive index, there is no phase change upon reflection.

Example 3 A Colored Thin Film of Gasoline

A thin film of gasoline floats on a puddle of water. Sunlight falls perpendicularly on the film and reflects into your eyes. The film has a yellow hue because destructive interference eliminates the color of blue (469 nm) from the reflected light. The refractive indices of the blue light in gasoline and water are 1.40 and 1.33. Determine the minimum non-zero thickness of the film.



27.3 Thin Film Interference



$$2t + \frac{1}{2} \lambda_{\text{film}} = \frac{1}{2} \lambda_{\text{film}}, \frac{3}{2} \lambda_{\text{film}}, \frac{5}{2} \lambda_{\text{film}}$$

$$t = \frac{m \lambda_{\text{film}}}{2}$$

$$t = \frac{m \lambda_{\text{film}}}{2} = \frac{(1)(469 \text{ nm}/1.40)}{2} = 168 \text{ nm}$$

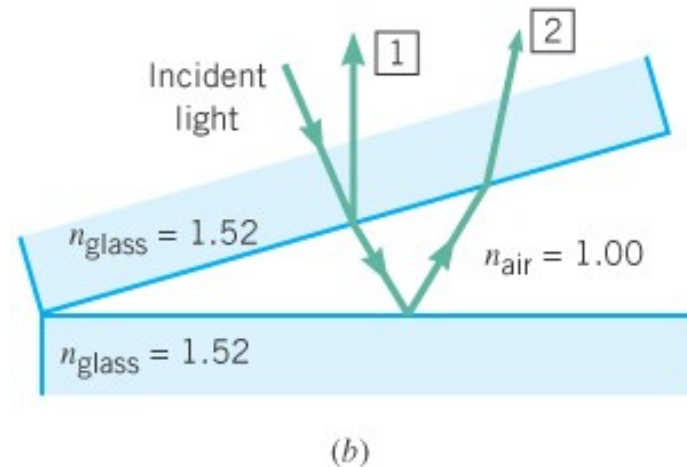
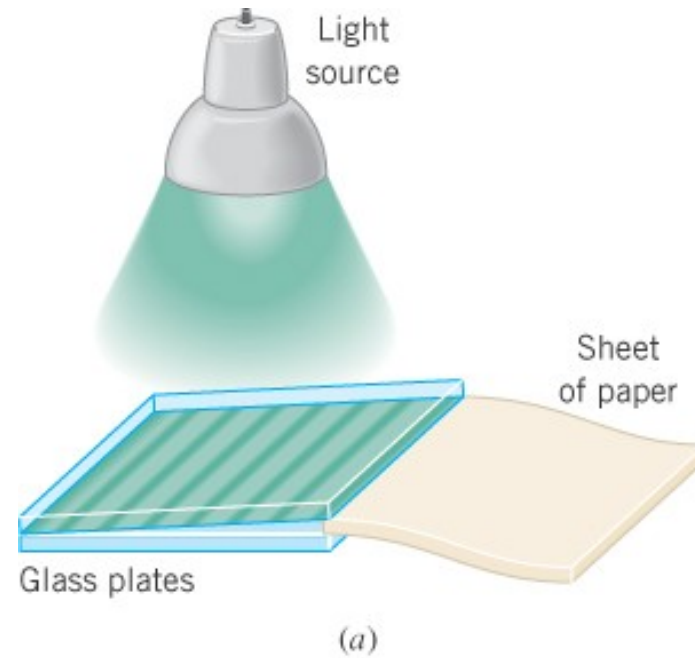
Conceptual Example 4 Multicolored Thin Films

Under natural conditions, thin films, like gasoline on water or like the soap bubble in the figure, have a multicolored appearance that often changes while you are watching them. Why are such films multicolored and why do they change with time?

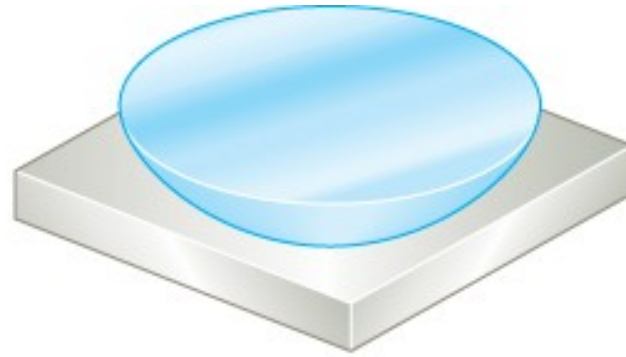


27.3 Thin Film Interference

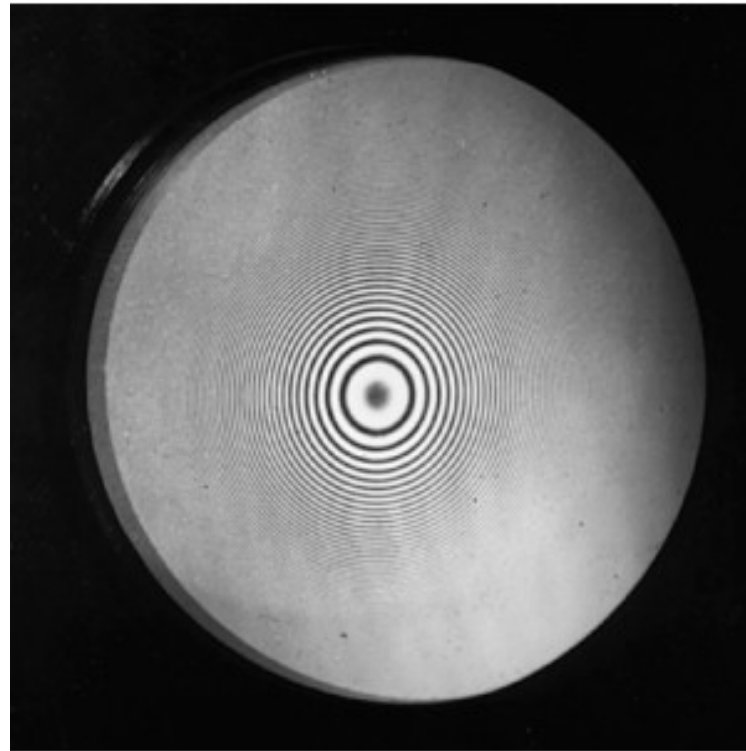
The wedge of air formed between two glass plates causes an interference pattern of alternating dark and bright fringes.



27.3 *Thin Film Interference*



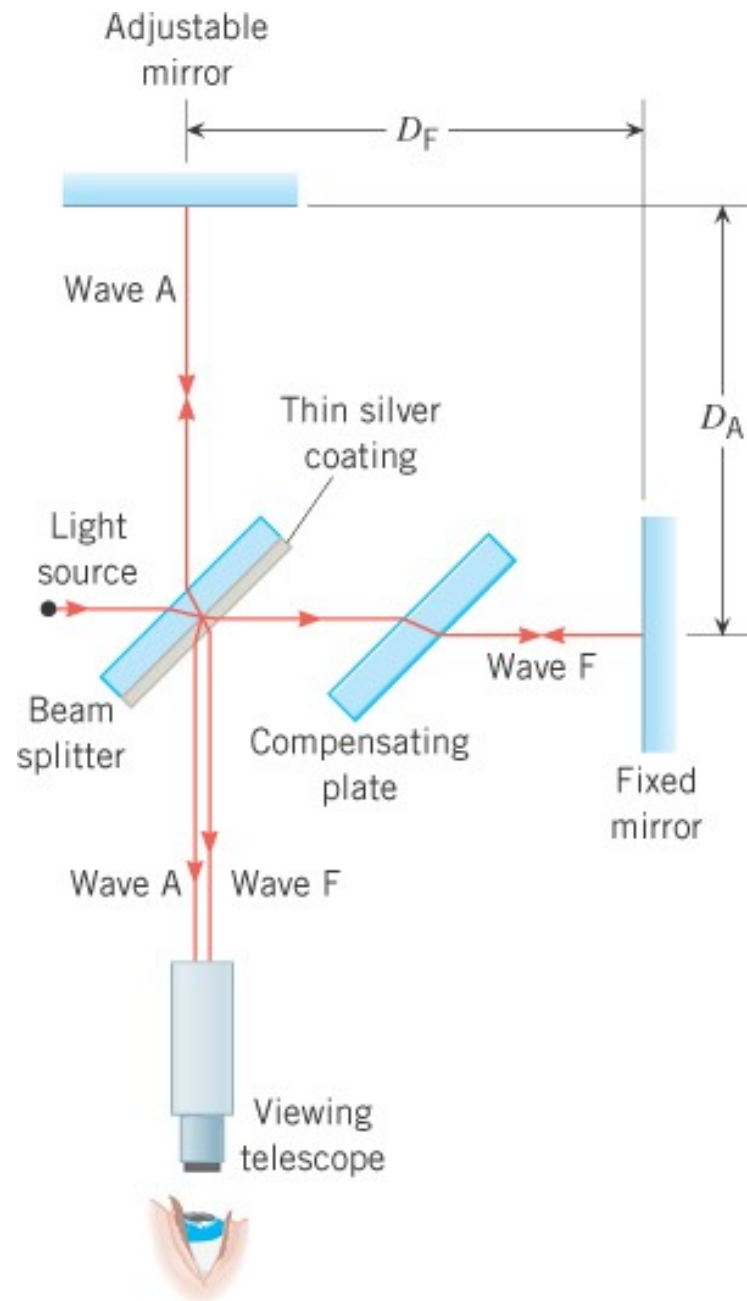
(a)



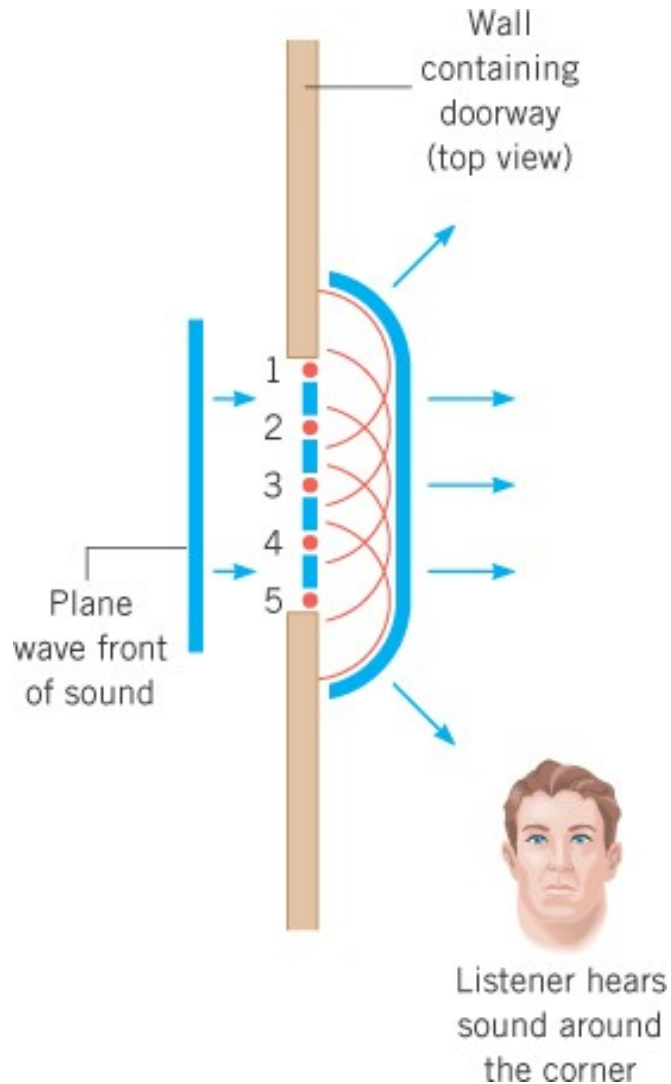
(b)

27.4 The Michelson Interferometer

A schematic drawing of a Michelson interferometer.



27.5 Diffraction

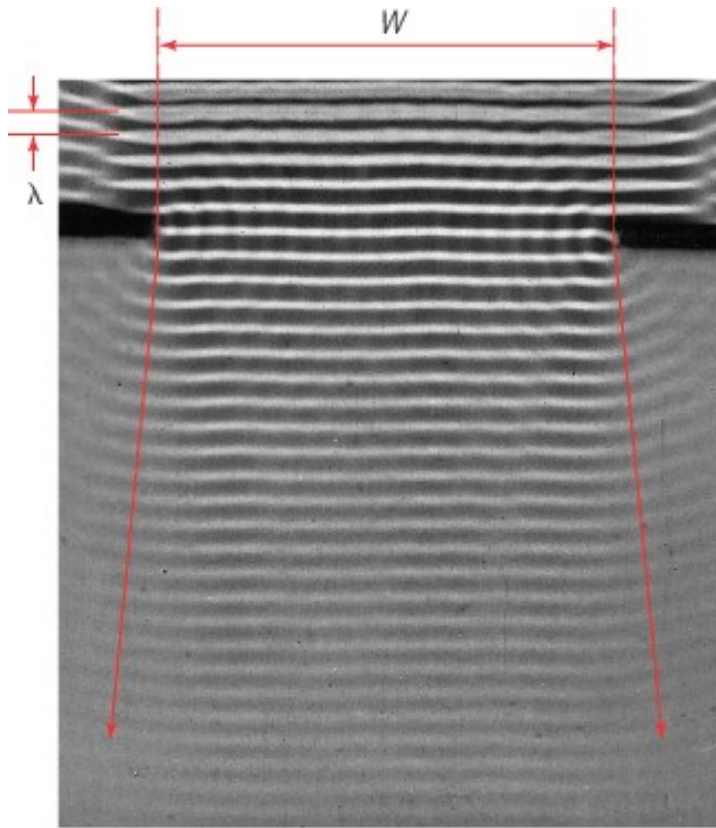


Diffraction is the bending of waves around obstacles or the edges of an opening.

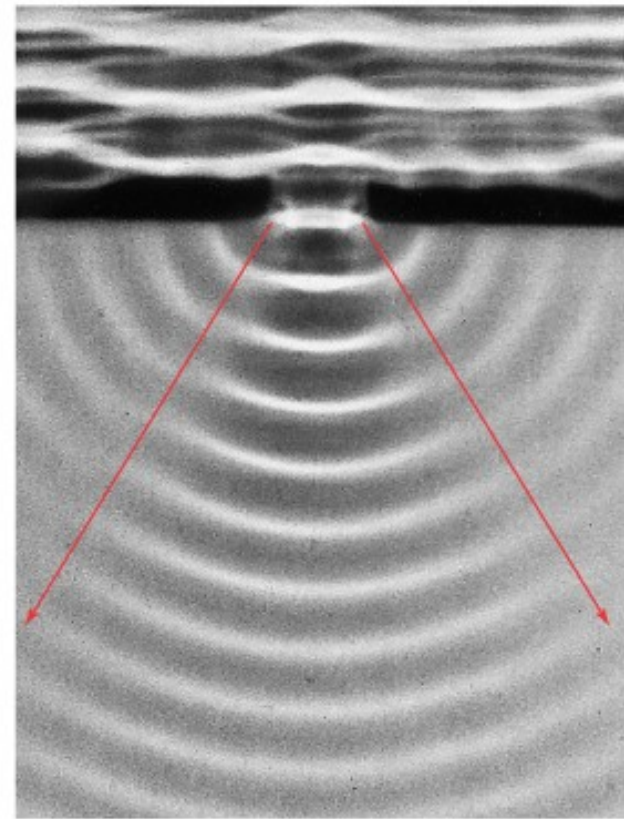
Huygens' principle

Every point on a wave front acts as a source of tiny wavelets that move forward with the same speed as the wave; the wave front at a latter instant is the surface that is tangent to the wavelets.

27.5 Diffraction



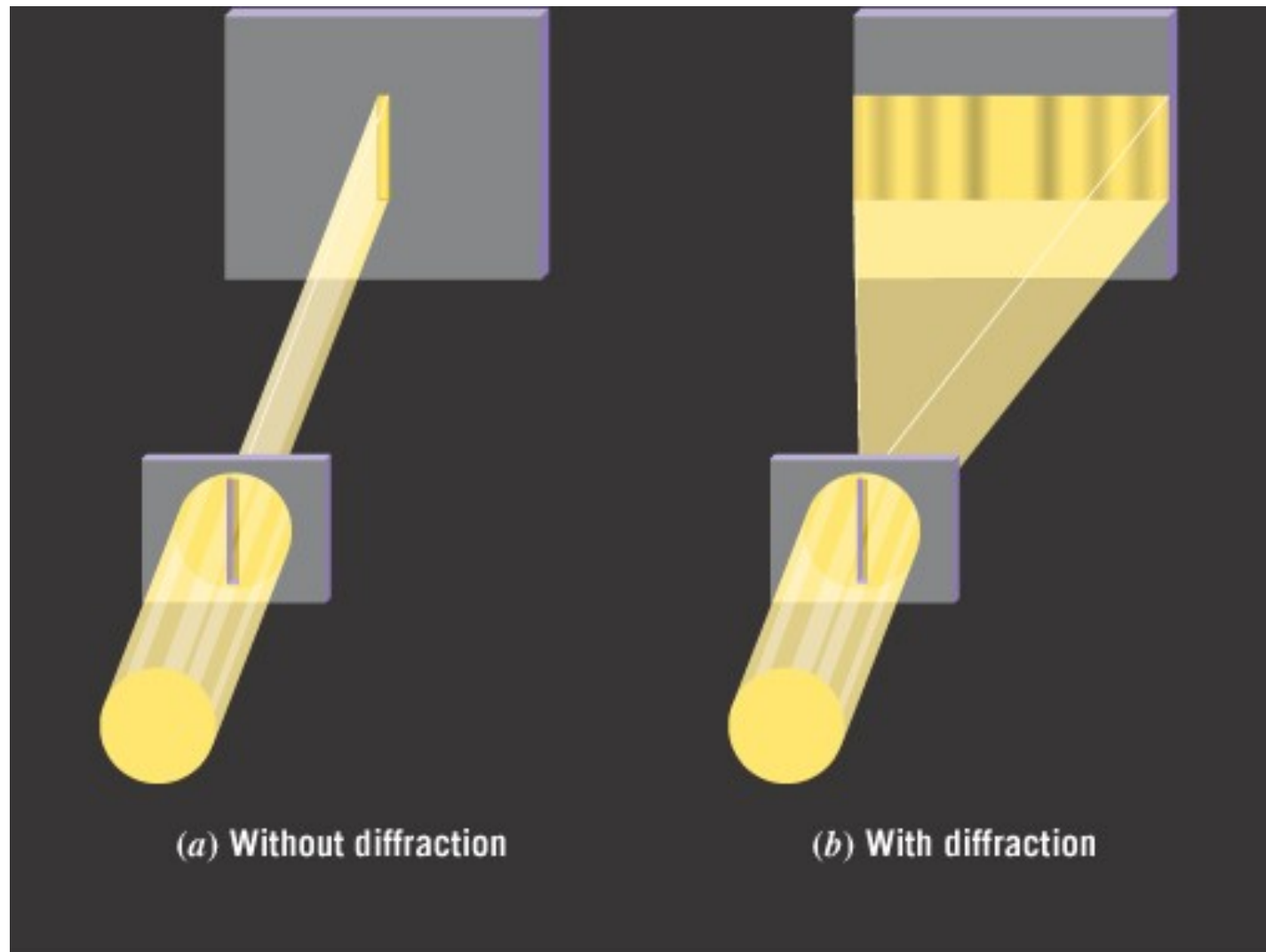
(a) Smaller value for λ/W ,
less diffraction.



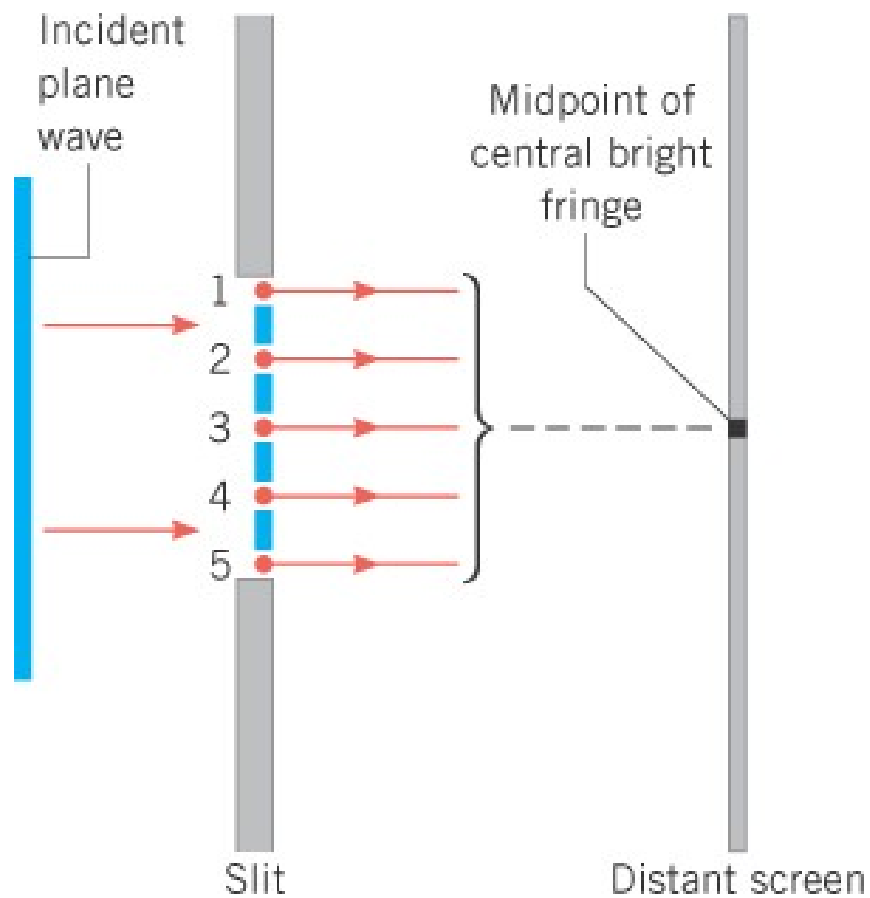
(b) Larger value for λ/W ,
more diffraction.

The extent of the diffraction increases as the ratio of the wavelength to the width of the opening increases.

27.5 Diffraction

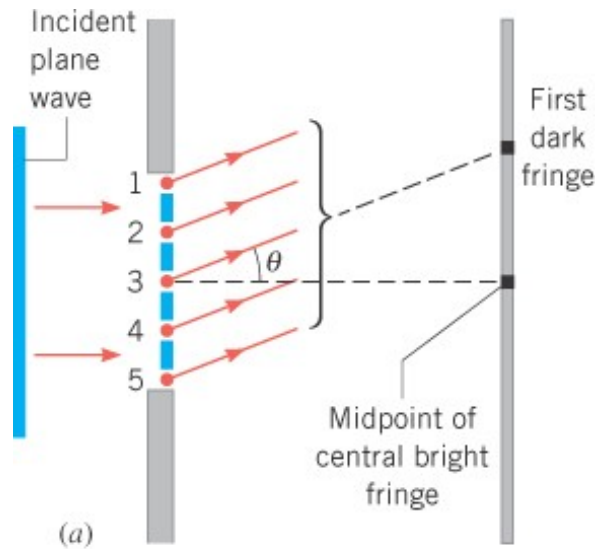


27.5 Diffraction

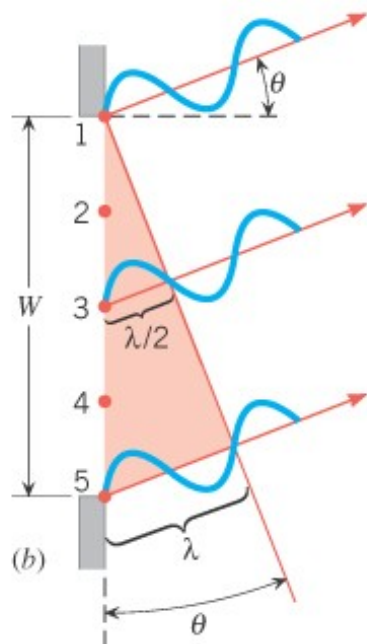


This top view shows five sources of Huygens' wavelets.

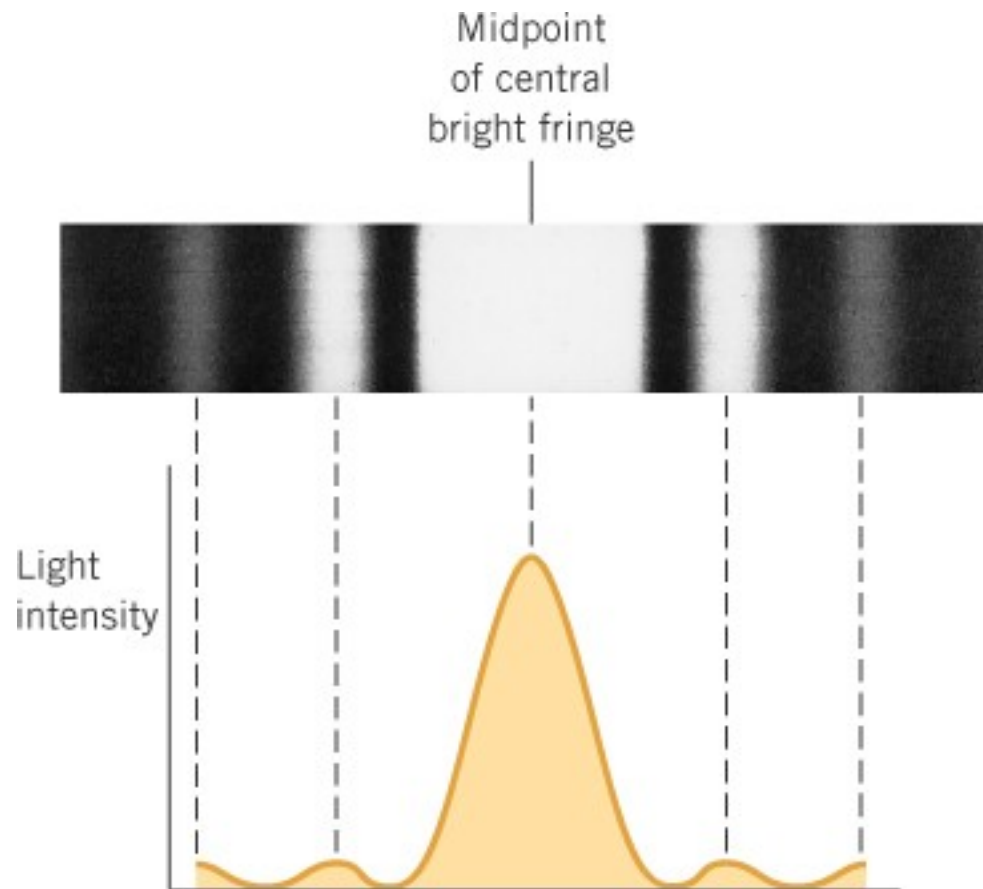
27.5 Diffraction



These drawings show how destructive interference leads to the first dark fringe on either side of the central bright fringe.



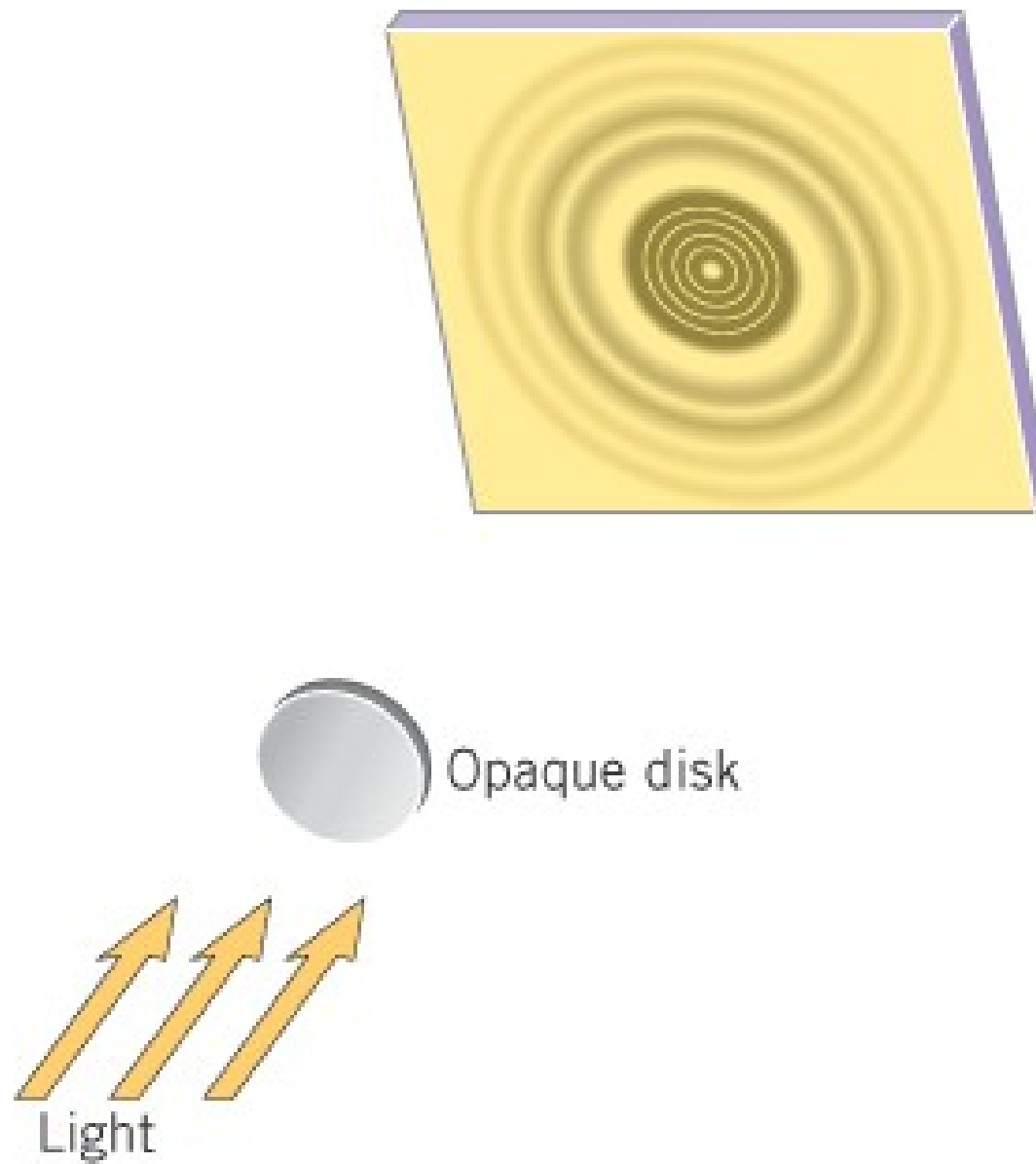
27.5 Diffraction



***Dark fringes for
single slit
diffraction***

$$\sin \theta = m \frac{\lambda}{W} \quad m = 1, 2, 3, \dots$$

27.5 Diffraction



27.6 Resolving Power



(a)



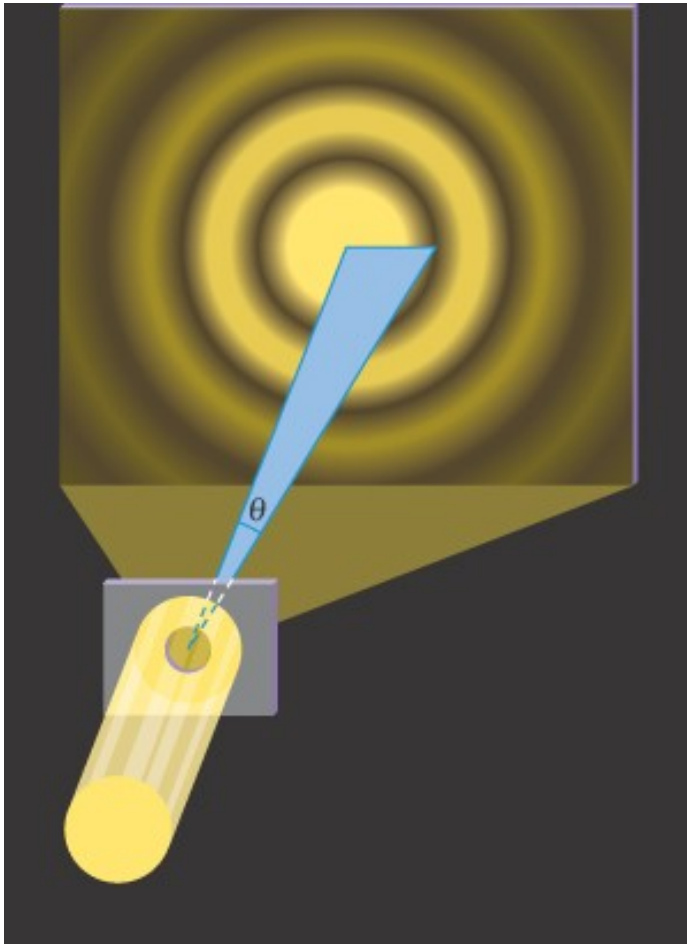
(b)



(c)

Three photographs of an automobile's headlights, taken at progressively greater distances.

27.6 Resolving Power



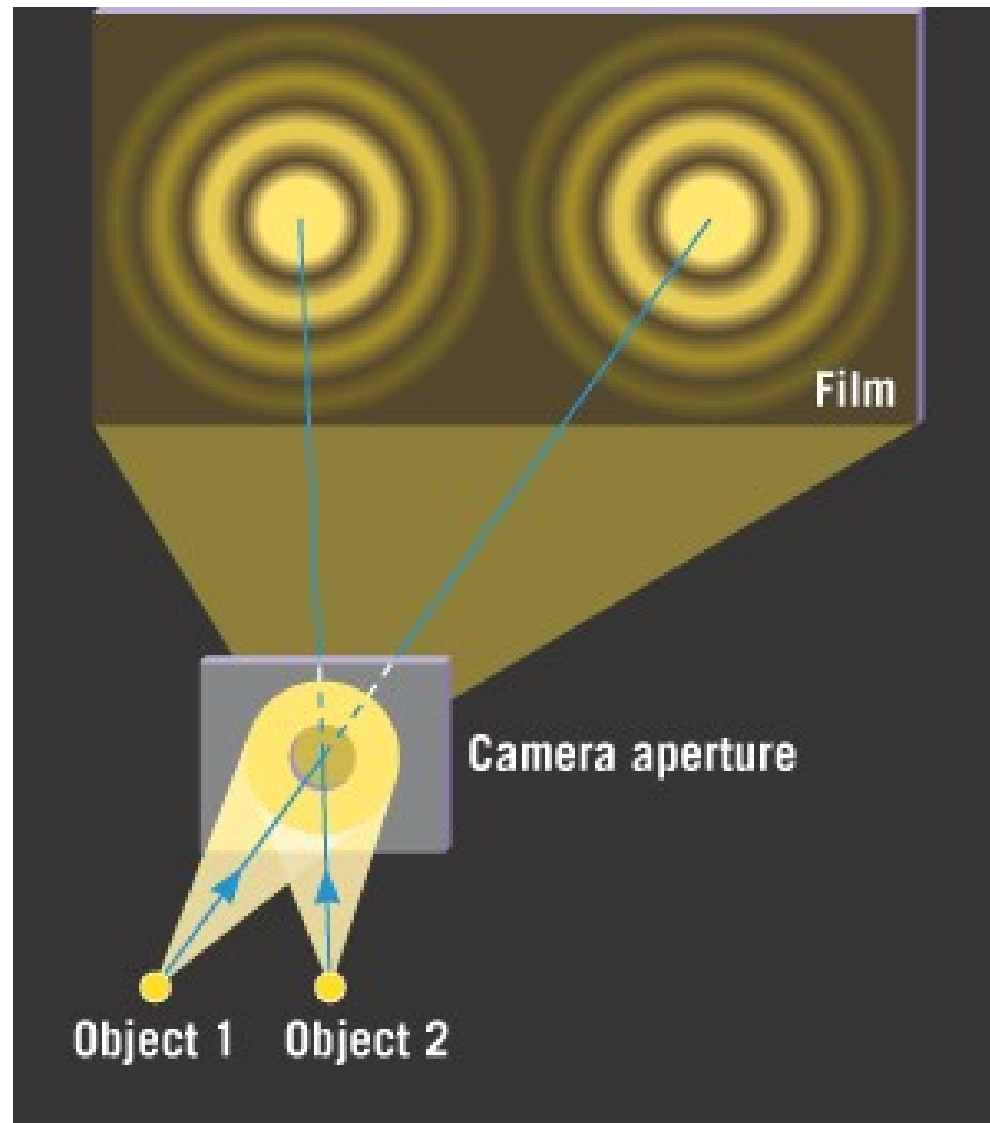
***First minimum of a
circular diffraction pattern***

$$\sin \theta = 1.22 \frac{\lambda}{D}$$

diameter of
hole



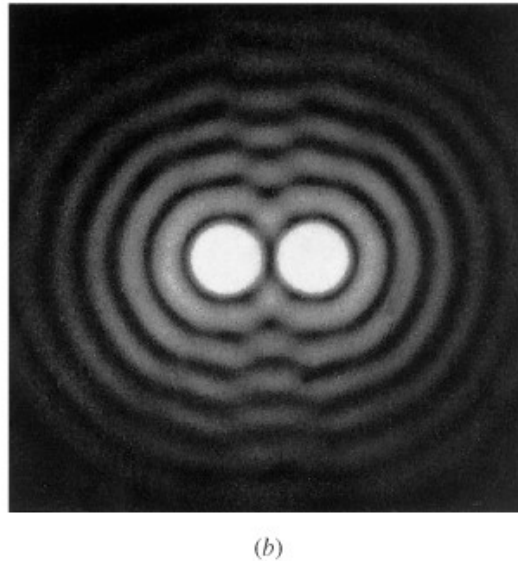
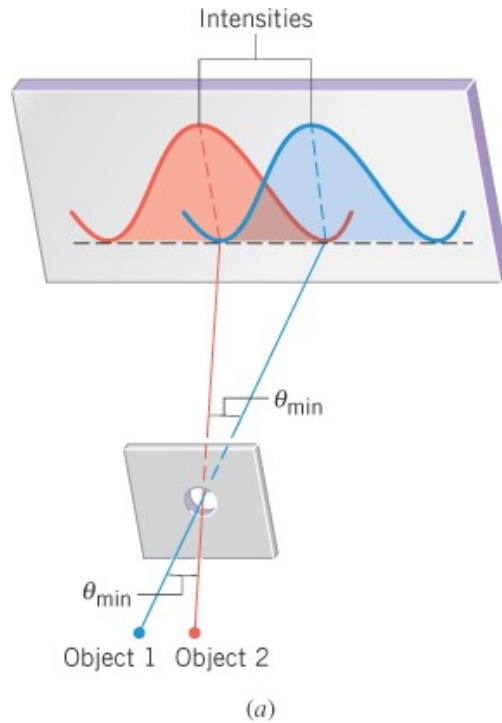
27.6 Resolving Power



27.6 Resolving Power

Rayleigh criterion

Two point objects are just resolved when the first dark fringe in the diffraction pattern of one falls directly on the central bright fringe in the diffraction pattern of the other.



$$\theta_{\min} \approx 1.22 \frac{\lambda}{D}$$

Conceptual Example 8 What You See is Not What You Get

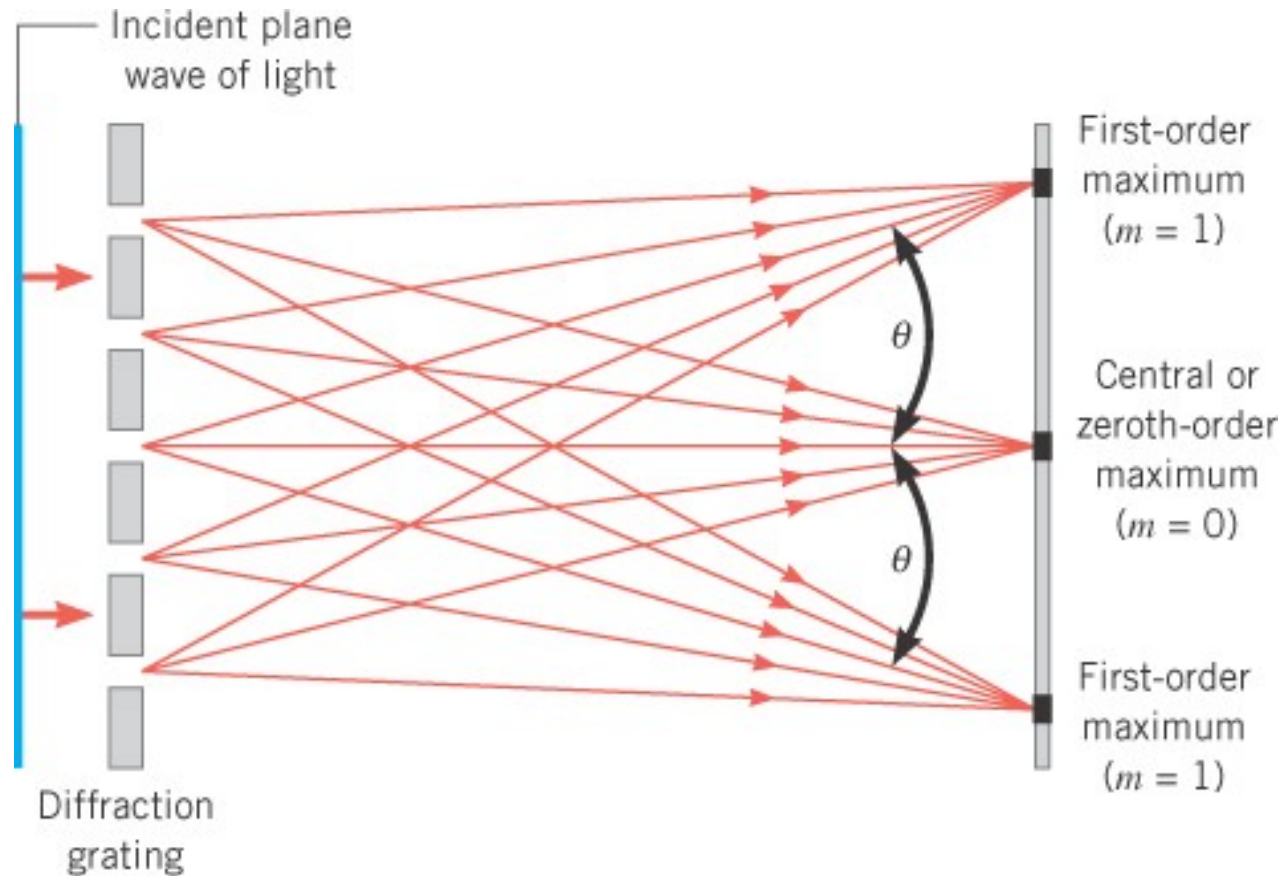
The French postimpressionist artist Georges Seurat developed a technique of painting in which dots of color are placed close together on the canvas. From sufficiently far away the individual dots are not distinguishable, and the images in the picture take on a more normal appearance.

Why does the camera resolve the dots, while his eyes do not?

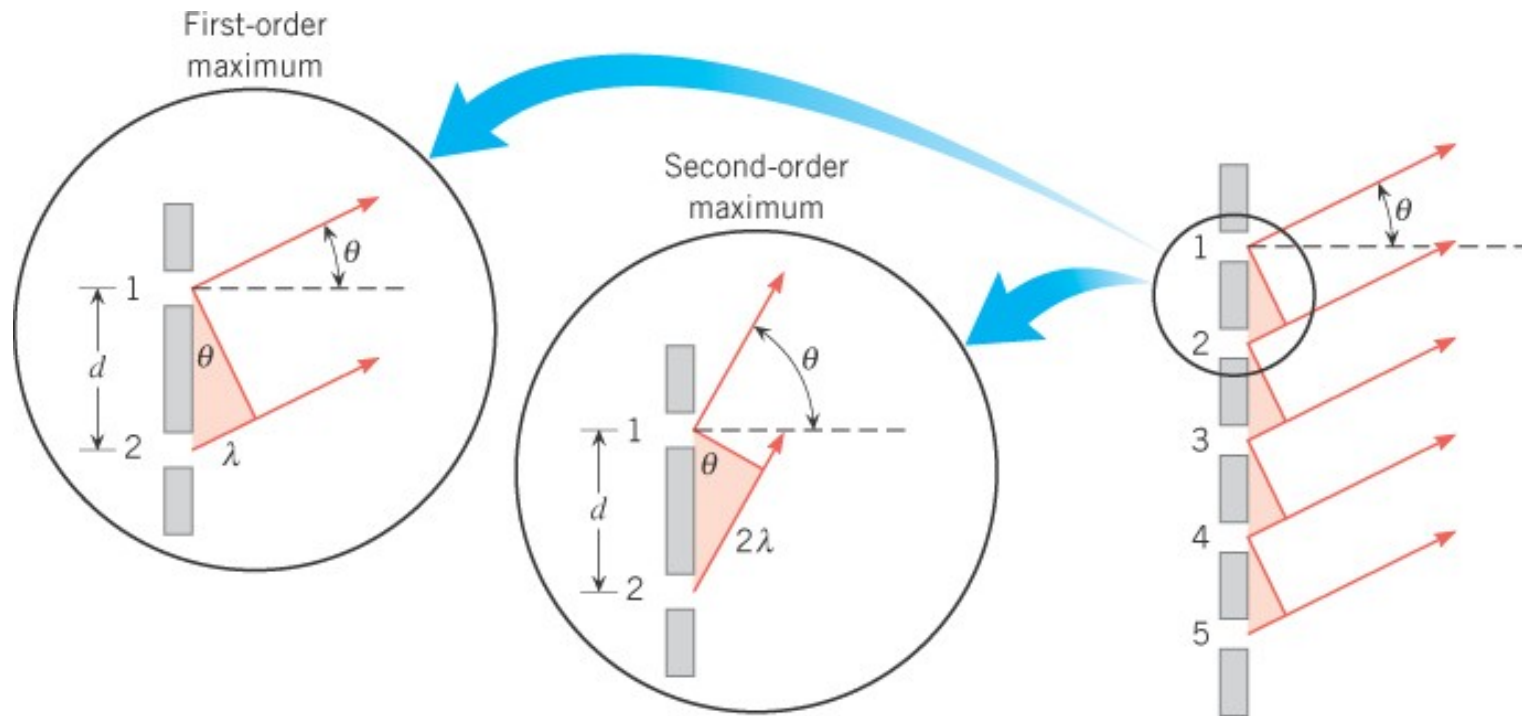


27.7 The Diffraction Grating

An arrangement consisting of a large number of closely spaced, parallel slits is called a **diffraction grating**.

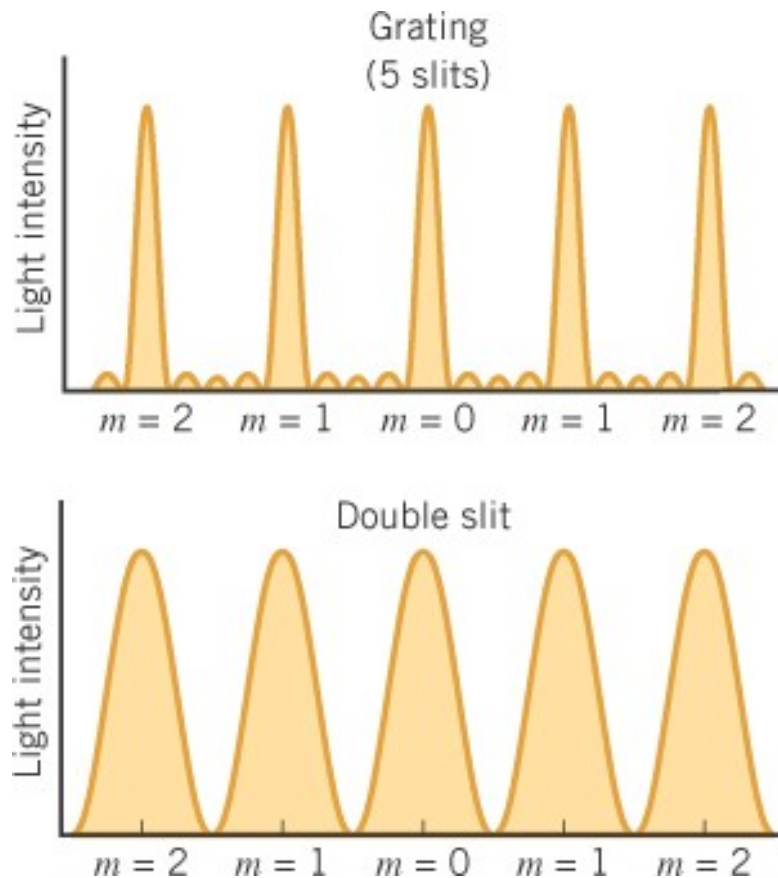


27.7 The Diffraction Grating



The conditions shown here lead to the first- and second-order intensity maxima in the diffraction pattern.

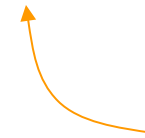
27.7 The Diffraction Grating



The bright fringes produced by a diffraction grating are much narrower than those produced by a double slit.

Principal maxima of a diffraction grating

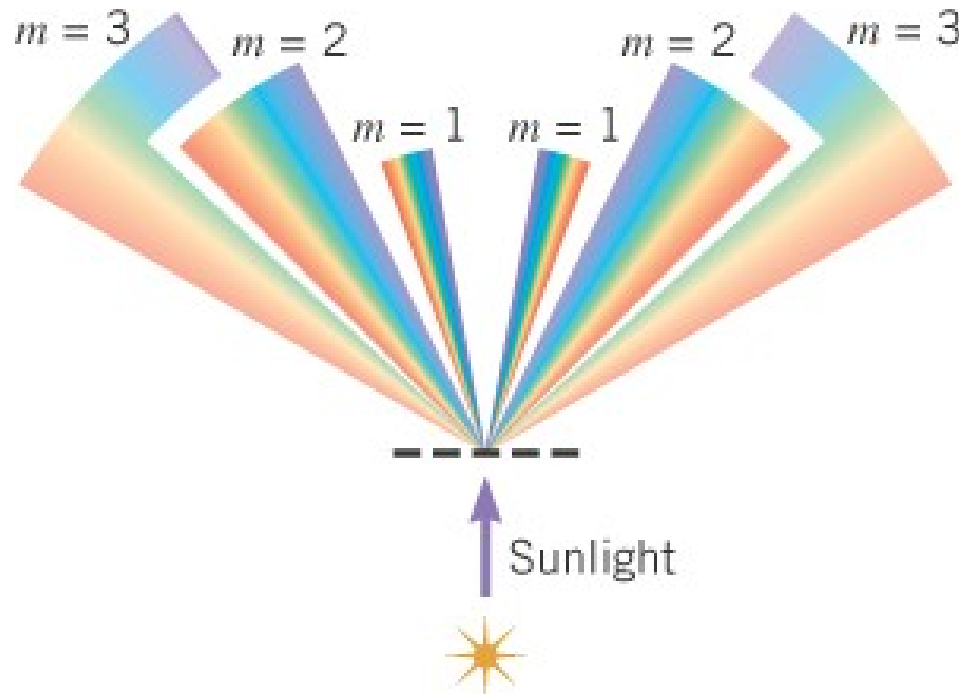
$$\sin \theta = m \frac{\lambda}{d} \quad m = 0, 1, 2, 3, \dots$$



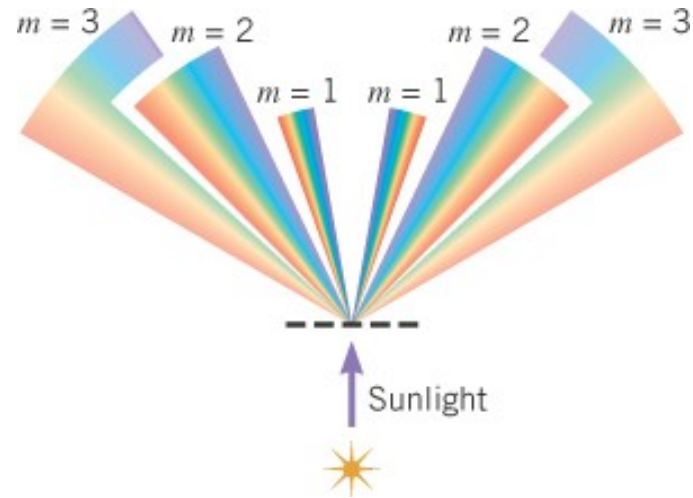
distance between
slits

Example 9 Separating Colors With a Diffraction Grating

A mixture of violet (410 nm) light and red (660 nm) light falls onto a grating that contains 1.0×10^4 lines/cm. For each wavelength, find the angle that locates the first-order maximum.



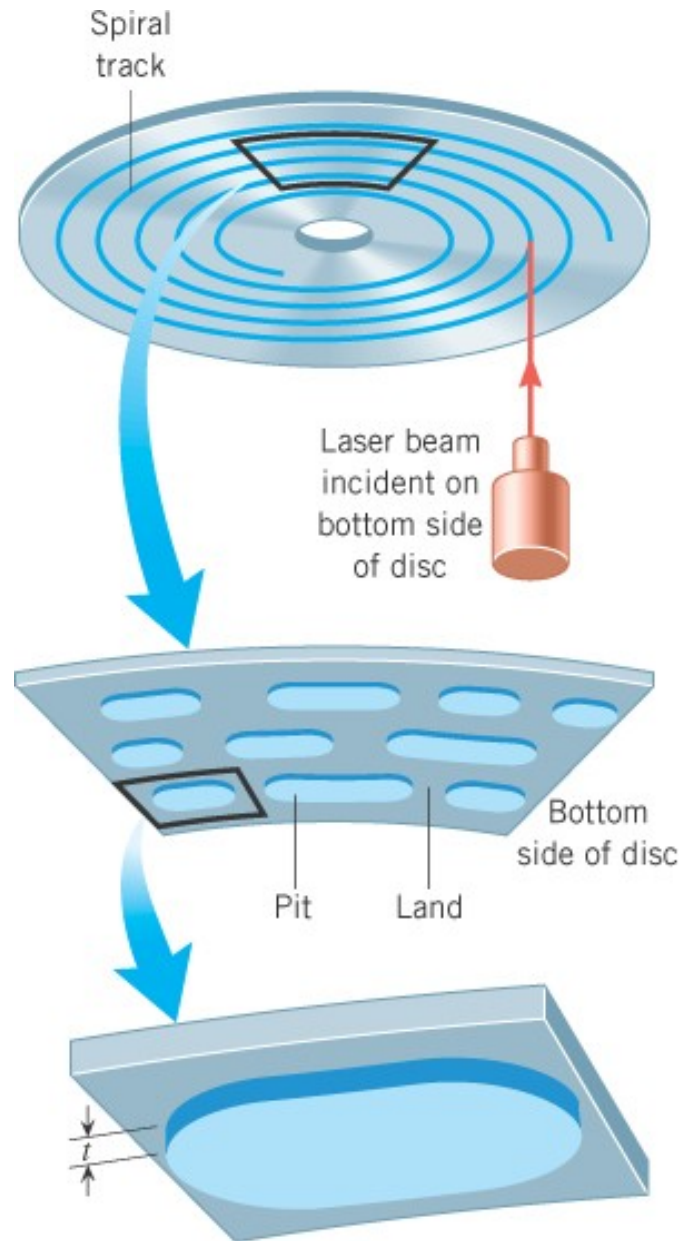
27.7 The Diffraction Grating



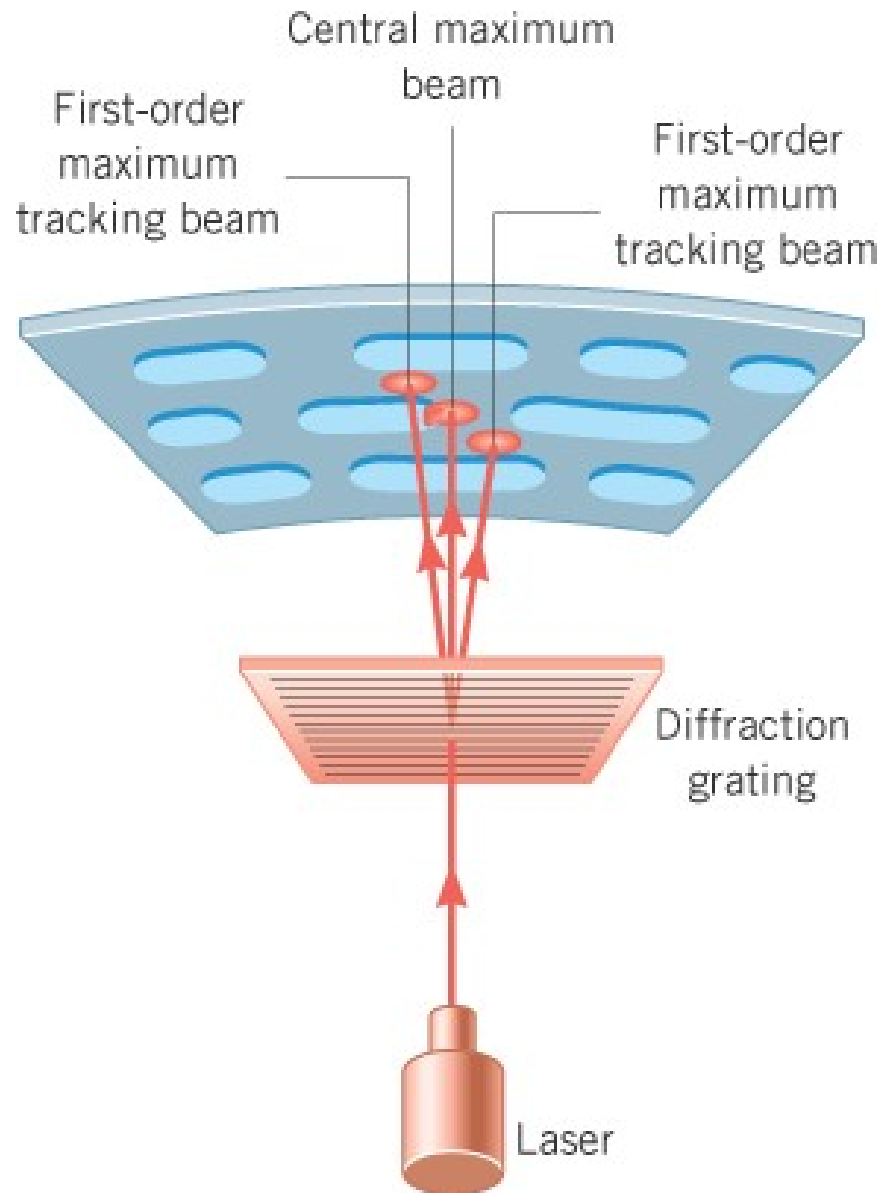
$$\theta = \sin^{-1}\left(\frac{\lambda_{\text{violet}}}{d}\right) = \sin^{-1}\left(\frac{410 \times 10^{-9} \text{ m}}{1.0 \times 10^{-6} \text{ m}}\right) = 24^\circ$$

$$\theta = \sin^{-1}\left(\frac{\lambda_{\text{red}}}{d}\right) = \sin^{-1}\left(\frac{660 \times 10^{-9} \text{ m}}{1.0 \times 10^{-6} \text{ m}}\right) = 41^\circ$$

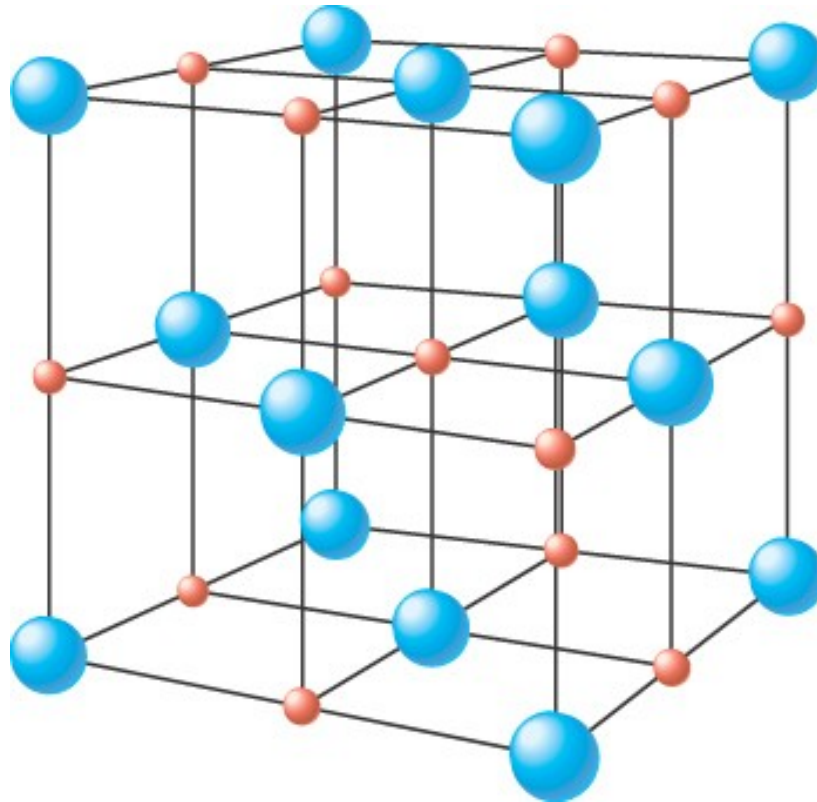
27.8 Compact Discs, Digital Video Discs, and the Use of Interference



27.8 Compact Discs, Digital Video Discs, and the Use of Interference



27.9 X-Ray Diffraction



27.9 X-Ray Diffraction

